Ep #21 | WTF is Longevity? | Nikhil ft. Nithin Kamath, Bryan Johnson, Prashanth, Jitendra & Seema

[Speaker 3] (0:00 - 0:14)

Okay, so what I want for today, A, to figure out what to put in my body, what not to put in my body. The second question I want answered, in this realm of health, where is the opportunity to start a business and what business should they start?

[Speaker 6] (0:30 - 0:47)

Ready? Rolling. Ready?

[Speaker 3] (0:48 - 1:02)

Okay, full house today. Thank you, all four of you for doing this. Brian, welcome to India for the first time.

What is the one thing you're looking at more than anything in India right now?

[Speaker 2] (1:02 - 1:03)

Air quality.

[Speaker 3] (1:07 - 1:08)

How bad is it?

[Speaker 2] (1:08 - 1:09)

I can't really see you over there.

[Speaker 3] (1:11 - 1:19)

This is Mumbai, we're mostly from Bangalore. So what does that mean? So what is the number right now?

[Speaker 2] (1:20 - 1:54)

The PM 2.5, I think it's 130, 140, which is over 10x what is reasonable, what is appropriate. So what is it doing to me? These contaminants wreak havoc on the body.

It can create neurological dysfunction, it can cause asthma, it can cause lung irritation, it's whole body damage. It gets into tissues, it lodges in, it's very hard to remove it. So it's not like something the body just cleans it out, it's very hard to remove.

So you really don't want to mess with these airborne contaminants.

[Speaker 3] (1:55 - 2:01)

So if I were to pick a city to live in, how big a role would air quality play in choosing which one?

[Speaker 2] (2:02 - 2:03)

Close to top three.

[Speaker 3] (2:04 - 2:08)

What would be the other options? What would be top one, two and three?

[Speaker 2] (2:08 - 2:09)

Not for health.

[Speaker 4] (2:10 - 2:15)

I don't think that really... I mean, air quality, water quality, maybe?

[Speaker 2] (2:15 - 2:30)

Air, water and food. Brian, do you agree? I guess you are thinking through this because you can filter almost any water and get it to be pure.

With food, you can grow your own food. But with air, you have no control. You're in a system.

[Speaker 1] (2:31 - 3:03)

So there's this idea of, can you make at least your home a blue zone? You can't externally change anything. So it's usually EMF, some radiations at home.

And then there is the air, which can be largely purified if you're in a very closed environment. And toxins in water and toxins in your vegetables. If you can manage these on a continuous basis in your home, you can at least make a blue zone kind of environment at home.

[Speaker 3] (3:03 - 3:07)

So is that what is working for blue zones? They have these three things going for it?

[Speaker 1] (3:07 - 4:31)

I think they, in addition to that, they define two other important aspects, right? One is how active they are through their daily lives. And the second aspect that they emphasize is community, right?

And how much of connectedness do they kind of... is imbibed into their way of living. I think these are the two other...

other than food. See, personally, I'm not very convinced on the idea of blue zones. There's a lot of data out there which suggests that blue zones are created.

They don't exist. So classic case of cherry picking. And so what they have done is they have mapped certain groups of population and omitted the same subset of people from the same area, just to highlight that there are blue zones that exist.

So there's a lot of controversy around it also. So at this point, I'm not very convinced that there are actually blue zones. Do people exist that live longer than your average lifespan?

Certainly. Take example of Japan. There are certain places in Japan where they have, you know, even higher lifespan than the rest of Japan.

But are there specific blue zones around the world where there's some sort of a special magical diet or some sort of magical lifestyle? The data is not convincing enough.

[Speaker 3] (4:31 - 4:32)

Do you agree, Brian?

[Speaker 4] (4:34 - 4:37)

Is it okay to put a mask on? Yeah.

[Speaker 6] (4:38 - 4:38)

Yeah.

[Speaker 4] (4:38 - 4:54)

Okay. Is that okay? Yeah, sure.

What is the air quality in New York, for example? Is it, I mean, I never really bothered to look for it. But what is the air quality in New York?

It's good throughout the US.

[Speaker 2] (4:56 - 5:14)

Ah, thanks. Yes. Yeah, the US does a good job on air quality, generally.

But... Yes. Oh, yeah, actually.

Thank you. Let's see, test this out. You feel better?

That's good.

[Speaker 6] (5:14 - 5:18)

Thank you. Thank you for addressing my emotional needs. I feel so much better.

[Speaker 2] (5:20 - 5:21)

This will knock out 95%.

[Speaker 1] (5:21 - 5:22)

No, but just curious.

[Speaker 2] (5:23 - 5:36)

I mean, were you feeling something or this is more precautionary? I mean, I feel it. It burns my lungs.

It burns my throat. And then I also just know, like, these level of pollutants, it really does wreak havoc on the body.

[Speaker 3] (5:39 - 5:51)

Isn't this a tough way to live, Brian? Like you travel the world, you're always on the move. If you were to worry about everything that can harm you, is it easy to live like that?

[Speaker 2] (5:51 - 7:08)

You know what, in many ways, it's easier. Because it makes decisions for you. So...

When you say it, what is it? Like the algorithm. So like if the objective is don't die, then it's just a systematic process of what inflicts damage upon the body.

And then how do you mitigate that damage? Like how do you block it? Or, you know, how do you mitigate it?

And so in that regard, the process predetermines what foods you eat, if you'll drink out of plastic water bottles or not, whether you pour glass, if you filter your water, if you wear a mask or not, if you go out and exercise. Like this morning, in the hotel this morning, we took this air purifier and went to the gym. They have this beautiful gym and I wanted to work out.

But we measure the air quality in the gym are really, really bad. And so it has this filtration. So the air from the outside is getting in.

So people are in there doing this really high exertion, deep breathing. And they're breathing in all these pollutants. They have no idea.

The thing is indoors. So it just really helps streamline decision-making across every aspect. And does the mask change that?

The decision-making? No.

[Speaker 3] (7:08 - 7:11)

The quality of air you're breathing significantly?

[Speaker 2] (7:11 - 7:26)

Yeah. So this will reduce 90. So the P at the PM 2.5. So let's look at this guy over here. It's 134. It will knock out 95% of it. Really?

Yeah. What kind of mask is it? N95.

Oh, like anything we use during COVID.

[Speaker 6] (7:27 - 7:29)

Yeah, those were the first. Because that's what we needed.

[Speaker 3] (7:29 - 7:29)

Yeah.

[Speaker 2] (7:30 - 7:33)

So 99 would be better. But we just had it. We had 95 on hand.

[Speaker 3] (7:33 - 7:35)

Is that what the 95 number is for?

[Speaker 2] (7:35 - 7:36)

It dribbles that pin.

[Speaker 3] (7:37 - 7:45)

Right. Brian, another question for you. Look at me.

Look at him. Look at him. And look at JC.

Who looks the healthiest? Take a minute.

[Speaker 2] (7:45 - 7:47)

Well, is there data to support? Do I...

[Speaker 3] (7:47 - 7:52)

No, no, no data. Who looks the healthiest? You're all different age groups.

[Speaker 2] (7:53 - 7:56)

Don't answer. You're dating me.

[Speaker 3] (7:58 - 8:03)

Come on. Who looks the healthiest? I'm not asking for number two, three, and four.

Number one.

[Speaker 2] (8:04 - 8:08)

Man. That puts me in a tough spot.

[Speaker 6] (8:08 - 8:09)

Yeah.

[Speaker 2] (8:09 - 8:17)

Hence the question. Upon what marker? So there's like, you know...

Overall health. So if you were to take an aggregate of all health markers.

[Speaker 3] (8:17 - 8:21)

Yeah. Not the markers, the optics of it. Who looks healthy?

[Speaker 2] (8:21 - 8:22)

Oh, who else looks healthy?

[Speaker 3] (8:24 - 8:26)

You can look at him too. He's quite fit.

[Speaker 2] (8:26 - 8:32)

I mean, most people make assessments of health based upon skin.

[Speaker 3] (8:32 - 8:35)

Right. And?

[Speaker 2] (8:40 - 8:53)

Nobody's going to feel bad. Well, we can't see your skin anymore. Yeah.

I mean, skin's like one of the biggest reveals of health, right? Like you can't fake skin.

[Speaker 3] (8:53 - 8:54)

Why is JC looking down?

[Speaker 2] (8:55 - 9:10)

I'm agreeing with what he's saying. He's right. So who is it?

We're all friends here. We're making friends. It's not good.

See, if I asked you to pick the worst, then you would not be making friends.

[Speaker 3] (9:10 - 9:13)

If you pick the best, you're making a best friend. Yeah.

[Speaker 2] (9:13 - 9:18)

At the end of this, when we're roasting each other, that's the time when we're making friends.

[Speaker 3] (9:20 - 9:34)

Okay. So maybe we start off where each of you can do two minutes about what you do in life and by health. Because I'm guessing Brian has not got context yet.

JC, would you like to go? All right, guys.

[Speaker 1] (9:34 - 10:02)

My name is Chitinder Choksi. You can call me JC. And yeah, we have a health and wellness company that is in the business of making people fitter and healthier.

And a fun thing. I didn't know, Brian, you were coming here. We bumped into each other, I think a couple of times.

So Akshay Summit, I was there. Yeah. You were there.

And I think Jeff from your team was in touch with us. Cool. Great to meet you.

Great to meet you.

[Speaker 3] (10:03 - 10:04)

Prashant?

[Speaker 1] (10:04 - 11:40)

Yeah, I'm Prashant Prakash. Otherwise, my day job is that of a VC, right? But in the last four or five years, I have kind of started this detour into looking at the idea of precision medicine.

So which is, you know, I mean, of course, there are names like Medicine 3.0 and other stuff, right? So what do you need for that? You need a strong science institution.

And that's where I co-founded Longevity India in Bangalore as part of the Indian Institute of Science, which is the premier institute. And so they will anchor all the science around longevity. And then BioPeak, which is the organization that will lean upon that science, but will put that science into practice, right?

So and what does that mean to put into practice? Can we be a bit more, a lot like what you do, and can you be a bit more data-driven and a bit more longitudinal in how you work with your clients? And number three, can you be more systemic and can you rely more upon systemic and cellular data and not just on blood biochemistry?

So I guess we're integrating a lot of this into, I guess, a new approach to how people in India can start viewing preventive, predictive and proactive health.

[Speaker 4] (11:41 - 12:08)

Yeah, I am Nitin. I'm the founder of a brokerage firm. It's called Zerodha.

And also run like a investment fund called Rainmatter. And through Rainmatter Health, we partner with startups like Fitter, and we invest in, and I'm also a customer of JC. So yeah, that's me.

And I'm better known as Nikhil's brother.

[Speaker 3] (12:11 - 12:18)

So tell him a little bit about health. So Nitin is like this obsessive workout guy.

[Speaker 2] (12:18 - 12:18)

Yeah.

[Speaker 3] (12:18 - 12:22)

I don't think he eats as many pills as you, but maybe like 60%.

[Speaker 2] (12:22 - 12:23)

Yeah.

[Speaker 3] (12:23 - 12:25)

I would say 30 tablets a day.

[Speaker 4] (12:27 - 12:29)

Yeah, maybe 20 to 30 tablets.

[Speaker 3] (12:29 - 12:50)

Not prescribed by me. And works out for a long time every day. And he had a health calamity earlier this year.

So he had a health problem. He had a stroke, even though he was very healthy. So I'll come back to you on that.

Like, you know, all this work that has gone into being healthy and don't die. Yeah. What would you do if you fell sick tomorrow?

[Speaker 2] (12:51 - 13:31)

So we, generally speaking, I'm really humble about the approach. Like we're probably in the very early stages of health and wellness. We probably know very little.

And we'll probably look back just in a few years time and think we're very primitive right now. And so those things happen. And it would be expected like those things are okay.

So we would just get back to work and figure out why it happened and what we can do about it. But we're definitely, I think, in the very, very early stages of medicine, like true anti-aging medicine. So yeah, I would just take it in stride.

[Speaker 3] (13:33 - 13:42)

This conversation is not me asking you guys questions. Everybody talks to each other, like always. So feel free.

Whoever wants to lead can start.

[Speaker 4] (13:44 - 14:49)

Yeah, I mean, I had the same question about last October, last November, I did Ironman. Like I was really at my fittest I was ever been. And January, I was met with a stroke.

It was so accidental. And it was strange that I had to go through everything that I've gone through this year. And this is like really the first time, I guess, I'm in front of a camera after the stroke.

And so, I mean, I want to ask as in, how have you, you know, even though it's early, you know, you just said that it's early in understanding human body, et cetera. But how have you, I mean, what would you do if you got a stroke or you got a heart attack? Are you measuring everything in your body to ensure that it doesn't happen?

[Speaker 2] (14:50 - 14:53)

Yeah, I mean, I'm the most measured person in human history.

[Speaker 6] (14:53 - 14:54)

Right.

[Speaker 2] (14:54 - 15:40)

Yeah. So everything we can measure, we do measure. Even if the measuring that thing doesn't have a high, like even if it's a new form of measurement, like DNA methylation became a new form of measurement a couple years ago.

And there wasn't much data to show that it was really, really useful. But we went all in. And I've been measuring my methylation patterns every week for the past couple years.

So more than anyone out there. And so, yeah, we take the opportunity to measure as much as we can, because we think the long-term data sets are interesting. So like if we can find patterns, like what would lead up to stroke?

Or what would lead up to heart attacks? And like, there's the common things you look at, you know, like the plaques or like, you know, other basic things like that. But we do try to piece the world together with data more than anything else.

[Speaker 1] (15:41 - 15:56)

So Brian, how much do you rely on metabolites and the multi-omics, right? So do you measure certain metabolites specifically as markers or indicators, right?

[Speaker 2] (15:56 - 16:07)

We do, yeah. We've also stored in a sample, a whole bunch of samples for future analysis. Right.

So yeah, metabolomics is like a new emergent field. Yeah.

[Speaker 3] (16:08 - 16:10)

I think for the audience, you have to give context with everything.

[Speaker 2] (16:11 - 16:11)

Yeah, yeah, fair.

[Speaker 3] (16:11 - 16:14)

If you say what is a metabolite, I think you need to explain what it is.

[Speaker 1] (16:14 - 16:36)

So metabolites are, so when you actually have these enzymatic reactions in your body. What is enzymatic? Okay, let's start with what happens when you take your food, right?

So when you take in your food, there is biotransformations that happen in your gut.

[Speaker 3] (16:36 - 16:37)

What is biotransformation?

[Speaker 1] (16:37 - 17:52)

Where your gut organisms actually convert that food into certain by-products, right? And those go through a series of enzymatic reactions. There are enzymes which are also released in your gut.

And from that, what you get is proteins and these other by-products, right? So and when you measure your metabolites, you actually measure, you know, if you think of this as a series of steps and in a reaction, right? Your metabolites kind of give you levels of different components in that equation.

So by being able to see not just the end product, which is in your blood, which is blood chemistry measures the end product, which may not always be indicative of what's happening in that process, right? Between when you consume your food to the final product. So being able to measure metabolites is a more fine-grained way of looking at what, it's a more precise way of looking at what's happening in the in-between steps.

[Speaker 3] (17:53 - 17:57)

Is that something that is happening at scale now? Is everybody testing for this?

[Speaker 1] (17:57 - 18:05)

Yeah, you can get it. So in India, you couldn't. And that's what we are setting up with Biopeak in a lab.

But in the US, I think there are labs now that can do it.

[Speaker 2] (18:05 - 19:02)

Yeah, we start with, that's a really advanced measurement. So we start with basic things like, if you're fat, you're a higher risk of dying. You know, if you have poor blood glucose control, that's also bad.

So you take like the things you can say with a high degree of confidence, and you start with those, and you work your way down to say where, like these kinds of things, it's emergent. This is like kind of interesting. We're not quite sure where and how and why.

But we really start with the basics of like, is getting sleep good or bad? Is getting exercise good or bad? Is eating, you know, like certain kinds of foods good or bad?

I think for most people, that's where they're at in life, is like they, there's, even on the basic things, there's so many questions, like you go out and search the world for health gurus, and everyone says something different. How do you even know what's right, what's wrong? It's very hard to parse.

So I think as a society, like we, we don't really know the basics of health anywhere.

[Speaker 3] (19:03 - 19:16)

Why don't you tell us a bit about how you got to be who you are? I think that's an interesting story too. Because you don't sound like a very traditional person or a normal, you know?

Yeah, yeah. Something happened there, so tell us.

[Speaker 2] (19:18 - 20:04)

Something traumatic must have happened. So yeah, I went to Ecuador when I was 19 years old. I was a missionary for my church, and it was my first time being exposed to poverty.

And I lived among extreme poverty for two years, and I just was so moved by the difference between what I had in the US and that poverty. What era was this, and how old were you? It was in Quito, I was 19 years old.

And so it's just like such a dramatic change as a kid to be in these poverty circumstances. And I thought, you know what, like... And back home, did you come from a rich household?

No, we were middle to low income. And you were Mormon? Yeah, yeah.

My single mom, five kids.

[Speaker 3] (20:05 - 20:14)

So you went from being born in a certain kind of family, being raised Mormon, to going to Ecuador when you were 19, and then?

[Speaker 2] (20:14 - 20:42)

Yeah, and then after that, I came back to the US after being in Ecuador, and I was trying to figure out what to do with life. And I just had this burning desire that I wanted to do something that would benefit the human race. Like, the people...

That would benefit? The human race. Right.

The people around me were really blue-collar workers, like they, you know, they weren't entrepreneurs, they weren't engineers, just blue-collar Americans.

[Speaker 3] (20:43 - 20:46)

Can I ask why? Like, why benefit the human race?

[Speaker 2] (20:46 - 21:33)

Because there was like two paths I saw. One is you could get a job, make money, and retire. Like, everyone around me had done that.

It's like, you just follow this life path. You don't dream big, you don't do different things, you just, you're a Mormon, you do life in a certain way, you retire. It's like, it's a very standard way of doing life.

Or, like, go out and carve your own path and do something unique. And I thought, that's cool. And, like, Ecuador was omnipresent on my mind.

And, like, these people, just by random luck, got born into this environment. And, you know, their life is pretty limiting. Like, it's hard for them to get out of that situation.

And it felt like that it might be a useful endeavor to do something that would benefit these people.

[Speaker 3] (21:34 - 21:42)

Right. So, you went to Ecuador, wanted to help human race for whatever reason, and then?

[Speaker 2] (21:42 - 21:51)

Yeah, and then, so the goal was, make a whole bunch of money by age 30, and then have liberty to spend money doing something. So, I started becoming, I wasn't good at anything. So, I thought.

[Speaker 3] (21:52 - 21:56)

Did you know you're going to spend the money to help others before you made the money?

[Speaker 2] (21:56 - 21:59)

Yes, that was the goal. Yeah, I told everyone around me.

[Speaker 3] (21:59 - 22:00)

That's quite rare.

[Speaker 2] (22:00 - 22:06)

Yeah, I've said, I'm going to make a whole bunch of money by 30. And then with that money, I'm going to do something interesting for the human race, like something useful.

[Speaker 3] (22:06 - 22:21)

It's almost against Maslow's law of hierarchies. I feel like you have to reach that level before you think of helping others. You're going to have to be self-sufficient first, right?

And then, how did you make the money?

[Speaker 2] (22:21 - 22:23)

I built Braintree Venmo, a payments company.

[Speaker 3] (22:24 - 22:24)

Okay.

[Speaker 2] (22:24 - 22:27)

And I sold that, and I made 300 million on the sale.

[Speaker 1] (22:28 - 22:31)

But this was, at what age was it when you got the money?

[Speaker 2] (22:31 - 22:46)

34. Yeah. Yeah, so then it was like, you know, okay, I said I would do this.

Now I'm here, and now am I going to deliver? Am I going to go buy a yacht, and get drunk, and do drugs, and you know, whatever? Did you?

No. For a bit? No.

Never?

[Speaker 4] (22:47 - 22:56)

Not even the medicated, the doctor prescribed, like the ketamine, and MDMA, and shrooms.

[Speaker 2] (22:56 - 23:06)

I mean, so when I was 34, yeah, in one year's time, I sold my company, I got a divorce, I left my religion, and I came out of depression. So all in one year's time.

[Speaker 3] (23:06 - 23:09)

Was the divorce because you got money?

[Speaker 2] (23:10 - 23:26)

It was a long time coming, and I think it was the catalyst of the event. And so it's just kind of like a reset in life, like it's rare you get these opportunities. And so it was like this genuine question, like what is existence?

Like wholesale, what is happening?

[Speaker 3] (23:26 - 23:34)

And also, what is money? I keep asking myself this question. But money is your ability to buy freedom?

Yes. Maybe through a divorce?

[Speaker 2] (23:35 - 24:10)

Entirely. I guess you got the confidence to redefine and say, yeah, now I can redefine myself. You'd be like, you know, when I made money, I wish somebody would have given me advice of like, let me just sit you down and tell you, money is a nightmare.

It is, like on so many levels. Explain. Like it warps all of your relationships.

It's like every person in your life now has, it's like you have three people in the relationship. You, your money, and the person. So you're like in this polyamorous relationship.

[Speaker 1] (24:10 - 24:14)

That's an interesting point, that everything gets colored now in all relationships.

[Speaker 2] (24:15 - 25:00)

It's just like something simple, like your niece has a birthday party and it's customary for like a $10 gift. It's nice. So your niece is like, oh, my rich uncle, he certainly could afford a $500 gift.

And so if you give a $10 gift, they're like, he's a bastard. Like, you know, like, what's he doing? Like, he has so much money.

Why can't he give me a $500 gift? Or you're out with friends for dinner and the check comes and everyone's like eyeballing each other. And it's like, why didn't you grab the check if you have the most amount of money?

So it's just like, it colors everything. And then it gets much more complicated with dating and stuff like that. But it really is like...

[Speaker 3] (25:01 - 25:09)

If you could have it another way, you don't make the money and don't get the importance that comes with money. Would you prefer that really?

[Speaker 2] (25:10 - 25:26)

I mean, I think honestly, in like parallel realities, if that were the case, I would rationalize it in my mind and make a case for it. Would you? I mean, I think in any situation, we're all going to make the argument that like whatever situation we're in now is the best situation.

Even when it's not or like objectively not.

[Speaker 3] (25:26 - 25:49)

But if you were to take yourself out of the picture and think in the third person, a man who made 300 million and is getting a lot of attention from women to his niece, to his friends and importance in that peer group versus somebody who didn't make the money and is another person. If you had to pick one binary choice, who would you pick? Whose role would you pick?

[Speaker 2] (25:50 - 25:54)

Yeah, for the coins. Like honestly, I think...

[Speaker 3] (25:54 - 25:57)

Nobody's seeing you make the choice. So you can't really prove a point. You're just making the choice.

[Speaker 1] (25:57 - 26:08)

But remember, Brian, the independence that you have now and your ability to kind of, you know, kind of define and redefine and do things your way.

[Speaker 6] (26:08 - 26:08)

Yeah.

[Speaker 2] (26:09 - 26:32)

If money, without money, would you still be able to do that? So like this is the thing where I guess what I'm... I'm basically, I'm calling bullshit on myself.

Like I don't care what I have to say about this question because whatever I say is total bullshit. I'm just making something up on the fly. I learned this when I was depressed.

Like my brain would say, you're depressed. Life is hopeless. You should kill yourself.

I mean, it just said that constantly.

[Speaker 3] (26:32 - 26:35)

What age was this? Was it after post-money, pre-money?

[Speaker 2] (26:35 - 27:12)

Pre, yeah, 24 to 34. So I just learned through that process to distrust my brain. And so if we talk about like, what are the effects of junk, you know, of like junk food on the body?

Like we can answer that with scientific analysis and say, here's the following effects on the following things. Great. If you ask this open-ended subjective question of like, what is a parallel life of you without money?

Which one do you prefer? Like, we're all just going to make shit up, but there's no way to base that in any kind of reality. So like our answers are basically a mirror of our own psychology.

They don't reflect truth. And so I guess what I'm saying is my mirror is what I would call bullshit on myself.

[Speaker 3] (27:13 - 27:13)

Or your mask.

[Speaker 2] (27:16 - 27:22)

I'd wear a mask and hide my attention. I'd hide behind a mask.

[Speaker 3] (27:23 - 27:32)

What happened after 34? After money. How did you stumble upon health as a thing or longevity or not die?

[Speaker 2] (27:38 - 27:44)

So one, I had started this process when I was depressed. Like how do you solve a really complicated problem in depression?

[Speaker 3] (27:44 - 27:51)

Sorry, I digress. But before we go there, you're 34, you made $300 million. Which city are you in?

[Speaker 2] (27:52 - 27:54)

That was Chicago. Chicago.

[Speaker 3] (27:54 - 27:58)

You're living in a nice apartment. You get divorced. What is the first thing you do?

[Speaker 2] (27:58 - 28:05)

Yeah, I moved to New York. Okay. And I learned that I love to dance.

[Speaker 3] (28:06 - 28:07)

Dance?

[Speaker 2] (28:07 - 28:21)

I've never danced in my entire life. I find out one night, I love it. And...

How did you find out? I was at a club with friends. Like we were at a business conference.

And they're like, hey, do you wanna go to the club? I'm like, what does that mean? I've never been to a club before.

[Speaker 3] (28:22 - 28:24)

You had never been to a club at 34?

[Speaker 2] (28:24 - 28:24)

No.

[Speaker 3] (28:24 - 28:29)

I get up until 19 because you were Mormon. But between the ages of 20 and 34?

[Speaker 2] (28:30 - 28:30)

No.

[Speaker 3] (28:31 - 28:31)

A bar?

[Speaker 2] (28:32 - 28:32)

No.

[Speaker 3] (28:33 - 28:40)

A restaurant? Yes. Why didn't you go to a bar?

[Speaker 2] (28:41 - 28:42)

In Mormon culture, you don't drink.

[Speaker 3] (28:42 - 28:44)

But you had left being Mormon, right?

[Speaker 2] (28:44 - 28:53)

Only at 34. I mean, I left before, but I was still in the family structure. But we just didn't hang out at bars.

It's not something we did. Or clubs or yeah.

[Speaker 3] (28:54 - 28:58)

Okay, so you moved to New York, went to a nightclub, loved.

[Speaker 2] (28:58 - 29:19)

I was actually in Morocco, I was at this business conference. And they're like, hey, we should go

to this club. And it was actually, it happened when I was trying to decide whether to get a divorce.

And I had a drink or two. I never really had alcohol. And so I felt a little buzzed.

Did you like it? I liked it. It was fun.

[Speaker 6] (29:19 - 29:22)

Yeah, it was cool. Yeah, it was nice.

[Speaker 2] (29:23 - 29:54)

And I just had this feeling that wash over me that it was that my kids will be better off if we were divorced than if I stayed in the marriage. Because the only reason I stayed in the marriage was for my kids. And I had this idea of a divorced household is damaging to kids.

And it's better to stay together. So I was like, can I stick this out? The kids get old enough and they move away.

And just like this never ending complicated question. And I had this realization that like they're better off if we do this.

[Speaker 3] (29:54 - 30:14)

And now you're a good looking guy. I mean, not everybody can see. Have you dated a lot of people post?

Post money, divorce, getting fitter, better looking? Yeah, yeah. What do you look for in a partner?

Because it's complicated, right? Like...

[Speaker 1] (30:14 - 30:29)

Very. I'm very curious who's checking his nighttime erections. Why is that the first thing you thought of?

Clearly you're asking him the wrong question. He's posting his butt naked pictures on social media.

[Speaker 2] (30:29 - 30:30)

What do you think that's for?

[Speaker 1] (30:31 - 30:32)

He's single and available.

[Speaker 2] (30:33 - 30:43)

Yeah. So actually next week, there's gonna be... Do you know Polly Market?

Yeah, no. Yeah, they're launching a market on my nighttime erections.

[Speaker 4] (30:44 - 30:51)

Wow. And how are they gonna... I mean, how are they gonna predict that?

[Speaker 2] (30:51 - 31:51)

Over, under is two hours and 12 minutes, my baseline measurement. And then I'll measure one time per month, last week of the month. But I think it's such a cool thing that this concept of...

So you have on one side, sperm counts going down, fertility is going down, birth rate is going down, sexual activity is going down. All these indicators of humans having sex and having babies, they're all going down. And so the nighttime erections is like a counter to these, of like sexual vitality is important.

And like sexual vitality, health, fertility, all that stuff. And so I'm glad that it broke through the zeitgeist, that now there's a market on it. So I think it really is a really good representation of society of like, we can bounce back with this vitality and not be beat down all these different markers, which is just not, I mean, the human race survives or dies based upon our ability to replicate.

[Speaker 1] (31:52 - 32:01)

So you think, is this one issue of why countries go through this population, why countries are going through this population decline?

[Speaker 2] (32:02 - 32:15)

I mean, there's probably a quite a few things driving it. But like, it's just, it's a fact we're having less sex, less, fewer babies. Just like, they're all down.

[Speaker 4] (32:16 - 32:30)

You're, I mean, I, you know, you kind of are very anal about taking any kind of supplements. No, not any kind of supplement. I'm simply saying like, take supplements which have enough evidence.

[Speaker 1] (32:30 - 32:58)

Like for example, I favor more supplements which you have deficiency of. Something that you

can observe in your blood or on adexa. Right.

But if you're talking about taking supplements that are working at a mitochondrial level, it doesn't have any proof. It doesn't have any validation. For example, if we talk about NAD precursors.

Today, we don't know if those precursors are actually getting absorbed in your system. Are they making their way to mitochondria?

[Speaker 3] (32:58 - 33:22)

Today, we don't even know what are the transporters that are going to transport NAD from your bloodstream across to cell membrane. Can you elaborate and say what is NAD a little bit? I've tried NAD once in Dubai on an IV.

And I felt so bad while I was taking it. It felt like my chest is heavy and it was so uncomfortable that I would never take it just for that reason. But what is NAD?

[Speaker 1] (33:22 - 34:00)

So in your body, there's a process which we refer to as metabolism. And your body produces something called ATP. So take whatever you want to, whether it's glucose or fatty acids.

And we'll exclude protein for now. The metabolism of the process end result is ATP. And in between, because the process is exothermic, but it also requires...

How do I explain it without getting too technical? No, I think see there is a whole oxidation process. Yes.

So you have a redox process. So NAD... Thank you.

So all the biochemical pathways in the body, they are coupled together.

[Speaker 3] (34:01 - 34:06)

So for everything, think idiot. Think I'm a 10-year-old idiot with no idea. Explain like that.

[Speaker 1] (34:07 - 36:03)

So your cells have nucleus. You are made up of trillions of cells. And your cells have different kinds of organelles.

One of them is mitochondria, right? So everybody knows that mitochondria is a powerhouse of the cell, right? So inside mitochondria, there's something called as NAD, NAD+.

And you have NADH, right? So these are basically electron donors and receivers. So when a chemical reaction has to occur, these donors are needed to complete that reaction, okay?

So they participate in something what we call as a redox reaction. And without these supplements, sometimes these redox reactions cannot happen. And so it reduces the mitochondrial efficiency.

Now I understood. Yeah, much better. But I think the added complication to that is you don't know if it's better to take NAD, NADH, right?

Or the precursors to that, right? So I think there's, you know, a lot of trials going. But see, NAD+, is needed very deep inside mitochondria.

So inside mitochondria, also you have inner membrane, which is where primarily the oxidation of fatty acids happen, which is where majority of your fat is getting burned. And there you need NAD, NAD+, okay? NADH.

So you can't basically channel an oral intake of NAD, NAD+, make it go through your digestive tract, and let's assume that it passes your digestive tract, you know, without any problems, goes into your bloodstream. There's a semi-permeable membrane, which it has to cross. And then once it's inside the cell, you can't tell that, hey, you have to go sit inside the inner membrane in the mitochondria.

There's no way. We can't navigate that. And there's no test.

[Speaker 3] (36:03 - 36:07)

So basically, should I have NAD or should I not? See, there's another problem, right?

[Speaker 1] (36:07 - 36:24)

You can't measure. Yeah, you can't measure. Because what you see in your blood is not necessarily what is there, let's say, in your muscle tissue, right?

Where you really need it. So just because you have NAD injected through your bloodstream, it might not be getting to the target.

[Speaker 2] (36:25 - 36:47)

Yes, what's your take on this? You take NAD? So yeah, there's little kids that have a ton of energy.

There's old people who are almost about to die. Little kids have a lot of this stuff. Old people have very little of this stuff.

And so the question is, if you give more of this stuff to old people, does it improve their health? So you take? I take it.

[Speaker 3] (36:47 - 36:52)

So there is one more, you know. And how do you take it? You take a NAD IV?

[Speaker 2] (36:52 - 36:58)

No, there's not good evidence. That's a good modality. I take it via capsule form.

I take NR, yeah.

[Speaker 3] (36:59 - 37:01)

And if I were to buy one, where would I get it?

[Speaker 2] (37:02 - 37:14)

Yes, I measure my levels. I do intracellular NAD. I do a blood spot card.

And we peg the levels to age 18. So there's like, the point is accurate. Like we don't, like does it have longevity benefits?

It's still an emergent science.

[Speaker 1] (37:14 - 37:34)

NR is available. Yeah, like you can just buy it online. Yeah, it's pretty popular.

There's an entire, NAD plus. NAD plus. So you get precursors.

So you get NMN. So NMN then translates. I take, you know, some, the precursors.

NMN you take? No, no, I don't take NMN. He doesn't take NMN.

[Speaker 6] (37:34 - 37:35)

NMN has a precursor.

[Speaker 3] (37:35 - 37:41)

He doesn't take anything. Hey, isn't NMN, okay, NMN capsules you get? Yeah.

[Speaker 1] (37:42 - 38:34)

So, you know, just to kind of, I mean, the membranes he was talking about, and you asked about this metabolites, right? So you can, slightly more, of course, this is advanced lab work. And this is what we're trying to do at Biopeak.

Can you measure some of those levels, which give you an indication? Is it, are you, where is the bottleneck? In that conversion process that he was talking about, you know, as you go through these, they call them, it's complicated, multiple complexes in your mitochondria, right?

As you go from complex one to two to three, there are different reactions that kind of take you from one complex. If there is insufficiency in a particular thing that can take you from one complex to the other complex, you have a dysfunction, right? Can you measure that?

And can you then target and see, what do you fix, right? That's, I think, a more precise way of doing it.

[Speaker 3] (38:34 - 38:44)

I think we're getting there. Can I ask another question? Maybe it helps, maybe it doesn't.

There isn't enough science. Can it hurt? I mean, look, that is a bad policy.

[Speaker 1] (38:44 - 38:50)

And that's not how you base science on. Because there's a lot of things. I'm not thinking like that.

[Speaker 3] (38:50 - 38:58)

I'm thinking as a supplement, because I've tried it. That's the entire supplement industry based on. Take it, it won't hurt.

Placebo.

[Speaker 6] (38:58 - 38:59)

You feel good.

[Speaker 3] (38:59 - 39:03)

But NAD plus or NMN can't really hurt you.

[Speaker 1] (39:03 - 39:28)

Look, most of the things that don't actively bind to, let's say, receptors, and they travel through your bloodstream, if they're not getting absorbed, they pass through your stool or pee. So, in a traditional manner, because NMN is recognizable by the body, whether the body needs it or not, depending on, it will just flush it out. So, most of the time you're eating it, it's going out.

Okay.

[Speaker 3] (39:29 - 39:34)

I think we'll come back to Brian now. 34 liquidity event, went to a nightclub in Morocco.

[Speaker 6] (39:35 - 39:37)

Had two drinks, loved it.

[Speaker 2] (39:37 - 39:48)

I like this transition. What did you do after two drinks? I made the conclusion, I'm going to get a divorce and the kids are going to be better off.

And then I moved my hips.

[Speaker 3] (39:48 - 39:50)

Was there something else that catalyzed that decision?

[Speaker 2] (39:50 - 40:13)

I just had been thinking about it for years and just finally hit. And I moved my hips at that very moment. And I just, it was like this pairing of freedom from all the things holding me captive.

And I spent the next, I guess ever since then, for it's been like 10 years now, I've just danced nonstop. Like I just love to dance.

[Speaker 3] (40:13 - 40:15)

How often do you get to dance?

[Speaker 2] (40:15 - 40:32)

I mean, when I was in New York, like multiple times a week. And where do you dance? We'd go to these big warehouse parties.

So like, you know, hundreds of people, great EDM. And we'd dance. We'd start at 12 or 1 and dance till 6 or 7 in the morning.

[Speaker 3] (40:32 - 40:33)

These are the kind of parties with no alcohol.

[Speaker 2] (40:33 - 40:37)

Everybody's in coffee or juice kind of thing. This is where everyone's doing psychedelics.

[Speaker 3] (40:38 - 40:39)

Do you do psychedelics?

[Speaker 2] (40:39 - 40:39)

I did.

[Speaker 3] (40:40 - 40:41)

What kind?

[Speaker 2] (40:42 - 40:43)

MDMA.

[Speaker 3] (40:44 - 41:00)

Have you researched the pluses and minuses of MDMA? I read somewhere that somebody is legalizing it for mental illness in America. Can you tell us a bit more about MDMA?

[Speaker 2] (41:00 - 41:26)

Yeah, there's a lot of interesting trials for veterans working on PTSD. You know, also psilocybin is very promising. So I'd say, yeah, psychedelics are having a resurgence in the US of interest where they got pretty buried by the government in the 70s, 80s.

And now I think that there's a lot of emerging evidence suggesting that they're interesting alts to pharma drugs.

[Speaker 3] (41:27 - 41:29)

In America, marijuana is legal to a large extent.

[Speaker 2] (41:30 - 41:30)

Yes.

[Speaker 3] (41:31 - 41:40)

Is that something you try or some? No. You think it has a lot of disadvantages as compared to an MDMA, for example?

[Speaker 2] (41:40 - 42:46)

Yeah. Yeah, I think marijuana is complicated. Complicated?

Like it just is, you know, like you take aspirin for a certain thing. You can do a psychedelic for a given therapy objective. You can, of course, use weed for CBD or can use THC or other versions

for like pain management or other things.

But it's just complicated. It has all kinds of other effects. And I think people sometimes confuse the specific application with the general enjoyment.

So it fills some broader category for them in their life. And then it could become more of a lifestyle thing. So I don't see many people succeeding with their objectives on marijuana.

Yeah. It's more, yeah. I don't know many high-functioning people who routinely use marijuana, maybe occasionally for enjoyment, but very few like actual regular practitioners.

[Speaker 3] (42:48 - 42:57)

Okay. So 34, beyond that, what happened? How did you arrive at this decision that what you're doing today is what will be the focus of your life?

[Speaker 2] (42:58 - 43:06)

It was a thought experiment that I was hanging out in the 25th century in a few hundred years from now. And imagining what they would say about this moment right now.

[Speaker 3] (43:07 - 43:11)

Is this zeroism and the dinners and... Yeah. Can you tell us?

I read a little bit about...

[Speaker 2] (43:11 - 43:52)

Yeah. So actually, if you guys are game, I'll run you through a few thought experiments. Yeah.

Okay, cool. Okay. So there's two rules here is everyone talks.

So everyone expresses an opinion. And number two is try to keep your answers short and snappy. Sure.

Because you will learn based upon other people's reactions to this. Okay. The first question for you is, if you had access to an algorithm that could give you the best physical and mental and spiritual health of your life, but in exchange for the access, you did what the algorithm said, you went to bed when it said, you exercise the way prescribed, you ate what it said.

Would you say yes? Or would you say no? Yes.

Yes is yes?

[Speaker 6] (43:52 - 43:53)

No. Okay.

[Speaker 2] (43:55 - 43:55)

Okay.

[Speaker 3] (43:56 - 44:04)

What if the algorithm told me to sleep at 8 p.m. every day and then wake up at 12 p.m.? I wouldn't do it. It comes in the way of my life.

[Speaker 2] (44:07 - 44:14)

Yeah. So maybe one or three yeses. Okay.

So no. Yes. Why?

Yes. I think it's fun.

[Speaker 1] (44:14 - 44:19)

You know, if there's a cheat sheet that can give you perpetual life, I think why not?

[Speaker 2] (44:20 - 44:21)

Okay. Yes. Why?

[Speaker 1] (44:21 - 44:32)

Yes. Because I think, I believe that, you know, when you have a lot of data computationally, I think algorithms make better decisions than what you can mentally process.

[Speaker 2] (44:32 - 44:33)

Okay. Yes. Why?

[Speaker 4] (44:34 - 44:45)

No, I mean, I generally like having like set of rules to follow. And, and if an algorithm is going to give me rules to follow, I just follow that.

[Speaker 3] (44:45 - 44:47)

What's the carrot? You live for perpetuity?

[Speaker 4] (44:48 - 44:50)

No, he's going to talk about the algorithm.

[Speaker 1] (44:50 - 44:53)

We are on the verge of figuring out algorithm. Maybe in future we'll figure out.

[Speaker 2] (44:53 - 45:10)

No, there's no carrot. There's no perpetuity. There's no carrot.

Yeah. No chance. The thought experiment gives you no promise.

The only thing that you know about this is you say, this is my best physical, mental, spiritual health. Outside of that, you don't know what you're doing. You don't know what it's saying.

You don't know anything about the best.

[Speaker 3] (45:11 - 45:17)

No, you won't. I consciously make choices which take me away from my best physical state ever possible.

[Speaker 6] (45:17 - 45:18)

Because you made one last night.

[Speaker 3] (45:19 - 45:24)

I don't see that changing. No, but the ultimate outcome, isn't that goal important?

[Speaker 1] (45:25 - 45:25)

If this kind of...

[Speaker 3] (45:25 - 45:39)

If that were more important, Prashant, we both wouldn't have gone and had a drink last night with myself. I feel like the quality of my life is just as important as long as my life. So what if the algorithm...

[Speaker 1] (45:41 - 45:48)

I mean, you could, you know, of course, get the data and then kind of make the choice.

[Speaker 3] (45:49 - 46:00)

But if I were to choose to listen to the algorithm or not, I won't listen to it all the time. If it goes

against the way I want to live my life, I won't. It's taking away my freedom.

[Speaker 1] (46:01 - 46:08)

But Nikhil, the bigger question is, if there is an algorithm, where is the algorithm? How far are we from it? And how easy or difficult is it?

[Speaker 3] (46:09 - 46:10)

So many... I think that's a digression though.

[Speaker 2] (46:12 - 46:41)

Okay, so that's question number one. Okay, question number two. Let's imagine we just...

The people who exist in the 25th century, they just watched this conversation. They heard me ask you the question and they heard your responses. What do they say about your morals, your values, your ethics, your characteristics of intelligence?

What do they say about you in this time and place?

[Speaker 1] (46:41 - 46:46)

Yeah, I think four of us would look very progressive and Nikhil would be wiping a pain in the ass.

[Speaker 3] (46:47 - 47:41)

Most evolved thinkers I have seen, I feel are not there today. They were there in the past. Because in the past, we didn't look at vocation or work like we do today.

There were professional philosophers and people who could devote a life to that. As you go higher up in the ladder of evolution, I think you ask more first principle questions of what is the point of life, get more nihilistic in nature, ask yourself whether you want to live in the first place or not. Much less, how do I extend if it is a miserable life a little bit longer and how do I make my life more miserable in order to achieve that?

So I'm assuming somebody in the 25th century is more evolved than I am. And I don't think evolution generally happens forward. It also can happen backward.

[Speaker 1] (47:42 - 48:01)

So it could also be possible, Nikhil, that at that point in time, people are more following these algorithmic directions because they again are focused on an outcome.

[Speaker 3] (48:01 - 48:03)

But he's not promising an outcome.

[Speaker 1] (48:03 - 48:18)

And what Nikhil is saying is that the progress doesn't happen linearly. Historically, if you look at the last couple of centuries, it seems like it has happened in a linear manner. But what he's saying is we are at a point where evolution and devolution, both seems quite plausible.

[Speaker 3] (48:18 - 48:32)

No, what I'm saying is evolution is not necessarily how we think evolution is going to be. Evolution could be different. People in the 25th century could be evolved in a different manner.

What do you think?

[Speaker 2] (48:33 - 48:41)

I'll tell you what I think after. So you guys think they would say what? You said that they would see a few of all.

[Speaker 1] (48:42 - 49:16)

I want to change my answer now. I think based on what Nikhil said, I do feel that my response was purely if we were progressing in a linear fashion. But now I'm not sure what they'll say.

Not too sure. I think they'll say that maybe they'll know a lot more about why certain things need to be done or not done. Right?

So they'll think that we probably were not really thoughtful in just saying that we'll follow the algorithm.

[Speaker 6] (49:16 - 49:17)

Mm-hmm. Yeah.

[Speaker 4] (49:18 - 49:29)

No, I would say I would think they will think that person's being stupid. This fellow saying no is maybe a stupid kind of a decision.

[Speaker 2] (49:29 - 49:30)

Yeah.

[Speaker 4] (49:31 - 49:41)

Who's saying no? No, no, I mean, in the future, the people looking back will think whoever said no to it as being stupid.

[Speaker 3] (49:42 - 50:25)

If the algorithm does not promise immortality in a way, I feel like there are many algorithms out there already today with no definitive fixed outcome in place. We are voluntarily choosing not to follow it, each and every one of us. I don't see why the 25th century people will not be like that.

I think human mentality is human mentality. Psychology of people will not change. All humans, in my opinion, outside of conditioning and conforming, pick pleasure.

Shorter-term pleasure is more valuable than middle-term pleasure. And that in turn is more valuable than far-off pleasure.

[Speaker 4] (50:25 - 50:32)

But do you think 25th century will think as drinking for pleasure?

[Speaker 3] (50:33 - 50:39)

I feel like we maximize pleasure more today than maybe the Greeks did 2000 years ago. So why will it change in 500 years?

[Speaker 1] (50:40 - 51:06)

There's a reason why it will change, right? So we have this idea of agentic copilots today that help us from the outside, right? And we have been kind of independent in that sense from external influences, so to speak, right?

But if you had that chip in your head, which is driving your thought process, right?

[Speaker 3] (51:06 - 51:15)

I don't think agents and copilots are thinking for us. No, no, they might start influencing your thinking, I'm saying.

[Speaker 1] (51:15 - 51:30)

Why not, right? That didn't exist the 10, 20, 30 million years before. But why would we think that, you know, in the future, the way we think and the way we get influenced will be the same way it happened.

[Speaker 3] (51:31 - 52:03)

Can I tell you why? Because if there is AGI and there is a chip which thinks better and not just

executes for us better or helps us, but things for all of us. Things for you specifically, let's say.

We'll soon get democratized and we all will have the same chip. We'll again go back to the innate cardinal for a man to distinguish himself from the peer. So you would do what is against the algorithm to find your identity versus following the algorithm.

But I think where that would evolve, I'm saying is undeterministic. Yeah, agree.

[Speaker 1] (52:03 - 52:15)

So in a sense, I'm saying, maybe my answer there would be, if I could rephrase is that, we just can't say because we don't know what somebody's thought process would be.

[Speaker 3] (52:15 - 52:16)

I'm changing your answers one by one.

[Speaker 1] (52:16 - 52:28)

Yeah. On the alcohol point, I have to point out that the Greeks and Romans didn't live long enough to see the consequences of drinking as much as the people of today are. Not alcohol.

[Speaker 3] (52:29 - 52:30)

Not alcohol. I'm not talking about alcohol.

[Speaker 4] (52:32 - 52:42)

I'm talking about… I was talking about alcohol. Saying in the 25th century, why would someone think you need alcohol for pleasure?

[Speaker 3] (52:42 - 52:50)

No. But would you let an algorithm tell you that you should do this and should not do this is the question.

[Speaker 1] (52:50 - 52:59)

But I think it's very possible that that could be a scenario is what I'm saying. Because we don't know how and what will control our thought process in that.

[Speaker 3] (52:59 - 53:02)

In my mind, it takes away individuality from… Maybe.

[Speaker 1] (53:02 - 53:06)

Maybe that's how we will become. I'm saying that it's a possibility.

[Speaker 2] (53:06 - 55:49)

So where we've been in these two thought experiments is the first question was purposefully provocative and open-ended, right? You're given this binary decision. Will you say yes or no to an algorithm?

You're given very little information. So you have to… In that process, you have to say, okay, complicated situation.

You went to your mind and you probed the database of ideas you have. How do I understand reality? You could pull any model you want and each of you chose a model of reality and said, here's my model of reality and my rationale, right?

But you basically show your models of reality through your responses. The second question, was a similar experience where you confronted a question with now a reflective tone where you're looking back on your model. So in the first one, you were very confident, like I'm going to do this for the following reasons.

This one, you're not quite so sure, right? Because we know from our historical perspective, we look back at previous generations, we think, ah, they weren't as advanced as we are. They didn't know things we know.

So we know more than them. We may be a bit more advanced than we imagined. It's possible that future generations will be more advanced than us.

And we think we might be humble a little bit to say like, maybe we are primitive in certain ways. So in two thought experiments, you go from a high degree of confidence of knowing to a less degree of confidence and more that you may not know. So just keep that in mind.

All right. Third thought experiment. So when I try to assess planet Earth right now, and I try to quiet my mind and acknowledge, try to identify what is really happening on the planet, I think there's only one thing going on.

We're giving birth to superintelligence. Nothing else matters besides that. We're giving birth to a form of intelligence that's going to change our existence unfathomably and in timeframes, which are hard for us to understand.

So as a species, we have this really pressing problem. What do you do when you give birth to superintelligence? Now, if you pose this question to China, they've got an answer.

If you pose a question to the U.S., they kind of have an answer, but the open AI has an answer and Anthropic has a different answer and Facebook's open model has a different answer. And if

you pose this question to Israel, they have ideas. If you ask an ideological group, what do you want to do with it?

They have certain ideas. So we have different people with different ideas all over the world, 8 billion of us, and they have different ideas on how they want to use superintelligence for war, for money, for status, for power, for sex. So a legitimate question when you're giving birth to an intelligence this powerful, what do we do?

And this is not a question of like, what ought we to do? This is like a legit question. We're a team, we're planning, what are we going to do in this moment?

[Speaker 1] (55:51 - 56:24)

The only thought that comes to me is that hopefully it's not about what, I mean, of course, a lot of better things can happen, but it does not lead to a dystopian situation and, you know, kind of existential human race kind of situation is what comes to mind for me.

[Speaker 6] (56:24 - 56:24)

Yeah.

[Speaker 1] (56:24 - 56:28)

Because you don't know where that superintelligence will actually end.

[Speaker 2] (56:28 - 56:29)

Yeah. So what do we do?

[Speaker 1] (56:30 - 56:54)

So you need to, in my mind, you know, I mean, have as much systems to govern the superintelligence and that the governance systems should evolve as in step with how fast the superintelligence evolves.

[Speaker 2] (56:54 - 56:55)

Whose governance systems?

[Speaker 1] (56:56 - 57:17)

Whatever that avoids that situation of humanity being, whatever, just destroyed at some level. Who do we call? So we'll have to figure that out, right?

We'll have to figure that out because there is everybody working on birthing that superintelligence.

[Speaker 2] (57:17 - 57:18)

Yeah.

[Speaker 1] (57:18 - 57:22)

Are there enough people working on saying, you know, how do you manage the superintelligence?

[Speaker 2] (57:22 - 57:23)

So what do you think we should do?

[Speaker 1] (57:24 - 57:46)

I think we should put enough effort on, I mean, you know, in terms of resources and, you know, putting people's capabilities behind, you know, this aspect that I mentioned. I think there has to be a little bit of refocus and shift.

[Speaker 2] (57:46 - 57:47)

Who does that?

[Speaker 1] (57:48 - 58:08)

I think in my mind, it's a collective global effort. And who's in charge? Whoever is, I mean, today, we are not nuking ourselves, right?

Everybody's, I mean, everybody can nuke everybody, but we're not nuking it and nuking each other. So I think the same systems that govern that.

[Speaker 2] (58:09 - 58:11)

So you think the status quo will take care of it?

[Speaker 1] (58:12 - 58:15)

No, no. I'm saying the status quo is a starting point.

[Speaker 2] (58:15 - 58:15)

Yeah.

[Speaker 1] (58:15 - 58:19)

Right? With the absence of anything else, I'm saying that's where I would start.

[Speaker 2] (58:19 - 58:21)

Okay. Others, what do you think we do?

[Speaker 4] (58:22 - 58:50)

I don't know. I don't know if we have control to do anything much. As in, the fact that there's US and China, I mean, how do you get them both to, you know, they're going to compete a way to birth this super intelligence, whatever.

So I don't know. I never thought about it, but I don't think we can do anything. Yeah.

[Speaker 1] (58:50 - 59:15)

Yeah, I think it's going to be a reactive response because it's growing in an unplanned manner. I mean, if people can't put a timeline around when AGI is going to be available, and we are so close to being there as some prominent people have pointed out. So if that's unplanned, I don't think anybody can plan a response now.

So it's going to be reactive response.

[Speaker 3] (59:16 - 59:23)

Yeah. I feel like we use it to increase productivity and solve problems wherever we can.

[Speaker 6] (59:23 - 59:23)

Yeah.

[Speaker 3] (59:24 - 59:31)

Including the problem of health. And like many innovations before it, it runs its course.

[Speaker 2] (59:31 - 59:38)

So similar pattern, like you just think like, so it's like two patterns, like we can't really do anything. You guys are like, kind of like, we'll figure it out.

[Speaker 3] (59:39 - 59:49)

I feel like everything evolves and we've not had much control with any of the evolution that has happened up until now. Why do we foresee controlling something that's happening tomorrow?

[Speaker 4] (59:50 - 1:00:15)

But up until now, it's not being a super-intelligent being. It's about, you know, things were under

your control. But you're talking about something, some super-intelligent being, which can think for its own self.

Which one becomes real then, right? Are human beings real or is super-intelligence real?

[Speaker 3] (1:00:16 - 1:00:19)

Everything is real. Everything that exists is real.

[Speaker 1] (1:00:19 - 1:00:29)

But, you know, yeah, everything that exists is real. But then which becomes, Which becomes organic is maybe the question.

[Speaker 3] (1:00:30 - 1:01:41)

The humans. I feel like I did a long conversation about this just recently with the meta Yann LeCun. His form of thinking is, LLMs, generative AI are nowhere close to AI.

They're a long, long way away. It's a completely different act of AI which can get into AGI. All kinds of intelligence that we have today is narrow.

A car being able to drive itself is narrow intelligence. A robot being able to operate on somebody is again a narrow kind of intelligence. If we have an intelligence which can combine many of these narrow intelligences right now, I feel like we use it.

I also feel like in this conversation, it's very important to define what we mean is intelligence. I don't think anybody has the right definition. They all have different definitions.

By virtue of having different definitions, they can't have the right one. So there's too many open variables in this. But if I were to like super impose my top-down thinking on all of those variables, I would say that it will evolve like everything else has evolved.

[Speaker 6] (1:01:41 - 1:01:41)

Yeah.

[Speaker 3] (1:01:42 - 1:01:49)

So like you said, somebody made a gun when we were punching each other. And today somebody has a gun, somebody does not, but the world is moving on.

[Speaker 2] (1:01:49 - 1:02:02)

We're all dead. So can I summarize this conversation, this part, this question and say we have no idea? Yeah.

As a species, we have no idea. I think that's a fair thing. Like not even a starting point.

[Speaker 1] (1:02:02 - 1:02:08)

Because I think what Nikhil said, we don't know what is super intelligence. I think even more fundamentally.

[Speaker 6] (1:02:09 - 1:02:09)

Yeah.

[Speaker 2] (1:02:09 - 1:05:56)

Right. What will be the form of that pure super intelligence? So can I pattern match that in the first question that I posed to you, if you had access to an algorithm, you went to your database and said, how do I understand the world?

You surveyed the models of your reality and you said, I'll choose that one and you mapped it onto the answer. This was the same pattern, right? I gave you a question that is inherently very complicated and very real.

You did the same thing. You went to your database of knowledge and said, how do I understand reality? And you mapped the model onto reality.

And you expressed the same degree of confidence, right? No one here said, I don't know. It was like, actually, I know.

And let me just tell you how this is going to work, right? So we mapped artificially some degree of confidence onto a question. We have no idea what to do as a species.

There's no plan. And so I want to suggest that we need a plan. And there's only one plan that I know of that actually can be equal to this moment.

And it's don't die. So don't die is a philosophical framework. It's an economic plan.

It's political. It's moral. It's social.

And it is don't die individually. Don't kill each other. Don't kill the planet.

And align AI with don't die. That don't die is the most played game by every human on this planet every second of every day. It is our zeroth order priority every second to not die.

Every breath we take of our metabolic energy is going to not die in this moment. That's true for all biology too. Most of biology.

And so don't die is the single thing that every human can agree upon. Above don't die, we branch

off into billions of different directions. And we have all kinds of different opinions about everything.

So don't die is what I've been trying to do for the past four years is I've tried to say what I want to become the most don't die person in human history. So how do you do that? You would say, I'm going to measure every way I can this biological, every organ, every biological process.

Take all the science you can apply all the science and say, what did you get? Look at the data and then repeat that. Measurement, science, data, measurement, science, data and do it again and again and again till you find that path of like what works, what doesn't work.

Now the same algorithm would work for planet Earth. You say, how could Earth be a sustainable biosphere for us as a species? You measure the Earth with billions of data points.

You say, I have a hypothesis on way good biosphere looks like. You run the experiment, you get data. Same thing, science, protocol, data.

And so it's the same thing again and again. Don't die is numerical in value. So just like I can measure my speed of aging or the age of an organ, you could say, what is the die score of a child's school lunch?

So in America, a common school lunch is a piece of pizza, chocolate milk, and canned vegetables. So like we know that's not a metabolically optimized meal necessarily. We could probably assign it a die score.

Like when a child consumes that, what is it doing for its health and wellness? You could also say, you can imagine this being a political situation where a politician is saying, I want to be elected because I give my constituents the best water, air, medical services, right? It's a political system built upon don't die.

It's numerical. And so I propose that don't die is the singular thing we focus on as a species, and specifically that we have power that by 2027, the same time when we think that these AI agents will be as good as our best AI researchers, that there's a billion of us around the world that are practicing don't die, economically, politically, religiously, morally, socially, ethically, all the above. What do you think about that?

[Speaker 3] (1:05:56 - 1:06:00)

Can I ask to what end? What is the point of life?

[Speaker 2] (1:06:02 - 1:06:07)

So that question is a luxury that one can ask when you're not dead.

[Speaker 3] (1:06:08 - 1:06:45)

Agreed. But if you were to take yours, I can't use the word soul, but take your mind out of your body. And if you have to compare and gauge two lives, one very happy, fulfilling 50-year lifespan, and one less fun 80-year lifespan, if your mind had to make a choice, which life should it pick?

What is the point of life? What is the meaning of it? So that question…

Don't die depends on that, in my opinion.

[Speaker 2] (1:06:45 - 1:07:10)

So that question has probably produced more murder than any question in history. And the question is basically, do you agree or disagree with this concept? Like, humans murder each other, when we disagree over concepts.

And so the point of don't die is don't die is the only thing in existence we agree upon. Everything else we branch and we kill each other.

[Speaker 3] (1:07:10 - 1:07:15)

When you think about it, you're not saying don't kill others, you're saying don't die.

[Speaker 2] (1:07:15 - 1:07:17)

Well, don't die, don't kill each other.

[Speaker 3] (1:07:17 - 1:07:24)

What if tomorrow it goes to a point, I'm just extrapolating here, that for you not to die, you have to like eat Prashant's heart or something like that.

[Speaker 2] (1:07:24 - 1:07:46)

What happens then? So the second rule is don't kill each other. So foreigners don't die.

Second is don't kill each other. Third is you can't destroy the planet. I'm safe, he has a second rule.

Second rule. And the fourth is, a line A, I always don't die. So like basically, we humans, we can't resolve conflict with death.

That's not an option. You can't kill each other to solve an argument.

[Speaker 3] (1:07:47 - 1:07:56)

What if you diminish the quality of his life? I'm just extrapolating because don't die in isolation, I think needs 10 other questions answered around it.

[Speaker 2] (1:07:56 - 1:09:22)

So it's a lot like democracy where the US was 13 colonies and they were ruled by the British. The US said, we want to do our own thing. We want to elect our own officials.

We want to govern ourselves like we want. At that point, they did not have democracy solved. They didn't know how to do states' rights, federal rights.

They didn't understand how to, what do you do with someone who's been committed for a crime? Can they still vote or not? Like there were thousands and thousands of questions that are unresolved.

The US is still fighting about democracy. 200 years later, still a constant debate. Don't die is the same way.

It's not that we have this solved ethically and morally. It's that you're pointing in the right direction. And so if you look at it from the 25th century, you really try to pull yourself out of the moment.

Don't die as a contemplation that when an intelligent species, when homo sapiens gives birth to super intelligence, what do you do? So like right now, we're in this very serious complication and we just had this conversation where it's like, we don't know. So if you go out and you say, hey, democracy, help us out or hey, capitalism, do you have any answers?

Or like, hey, Islam, hey, Christianity, like, hey, Buddhism, who can help us solve this puzzle? Like, what do we practically do? What is the problem you're trying to solve?

What do we do when you're giving birth to super intelligence? When you're giving birth to a form of intelligence that exceeds you in every capacity. Are you saying don't die in relation to that?

Exactly. I'm saying don't die is the practical, moral, ethical, social, philosophical, economic, political framework of what we do as a species when you give birth to super intelligence.

[Speaker 1] (1:09:22 - 1:09:33)

So is that just a different way of saying, you know, let's improve the quality of life of everybody on this planet?

[Speaker 2] (1:09:33 - 1:12:32)

How is it different from that? So it's both a negative statement and a positive statement, right? So a negative statement is that if the U.S., so whoever acquires super intelligence, they're probably not going to be, you know, they're probably according to current laws of society, which is ruthless brutality of power acquisition, right? Like that's what humans do. And so we do this because we value security and power and status. That's what we all do individually.

It's what we do collectively. So whoever acquires this technology is going to play by the rules of society to acquire those endpoints. So when you acquire this power, if you apply the same human rules of what I'm saying is, you have a possibility that leads you to a destruction of species.

So either AI can go wrong or humans can use it for the wrong purposes. And so what I'm saying is like in, when it's just humans playing the game and death is inevitable, like should I go to war, kill each other, create kingdoms, have someone else taking place in that kingdom, like do your thing. When you're giving birth to super intelligence and the map is changing for the species, you have to ask, is there any existing system that humans have that answers this equation?

Let me give you another example. So imagine you're hanging out with Homo erectus a million years ago. They have an ax in their hand.

That's their tool of choice. So they give you an idea of how either primitive or advanced they were. And you say, Homo erectus, tell me about the future of intelligence.

What are humans going to be like? Now, Homo erectus at that point may say, we're going to do, we're going to do more hunting. We're going to do more gathering.

We're going to travel to different parts of the world. But they can't tell you about the microscopic world of atoms. Or molecules.

They can't tell you about the electromagnetic spectra, which you can't see. They can't tell you about radios. They can't tell you about antibiotics, a pill you hold in your hand to swallow.

They have no models in their mind that inform them on any part of our reality. So the only thing a Homo erectus could say that would be intelligent is, I don't know. I have no models to state this.

Now, when we're giving birth to super intelligence, the question is, are we equal to Homo erectus? Are we basically as primitive relative to AI as Homo erectus is to our current state? And so that's the question for us is, do we know anything?

Which takes us back to our first question, our second question, and third question. This entire conversation is a juxtaposition of what we think we know, what we actually know, and trying to find the right balance between do we dare even express an opinion? And so don't die is basically saying, we have no idea what the future is offering.

We have no idea what's coming our way. The only thing we know, the only thing we can say with confidence is none of us want to die right now. That's it.

We can't say anything intelligent beyond that statement. Everything else is pure speculation.

[Speaker 1] (1:12:32 - 1:13:21)

Going back to Brian, what you're saying, I mean, does that even matter? That's fine. I mean, just this don't die thing when let's say you have two countries, let's say the US and China, like hypothetically, right?

So if they have all the AGI in the world and the rest of the world is pretty much. Basically what Prashant is saying that the philosophy of don't die has nothing to do with what's going to happen in the world. I mean, it seems good as a personal project.

It seems amazing that humans can improve their lifespan and increase, you know, the quality years in their life. But how does it all tie in with the future? I'm just correlating these two is where I'm trying to respond.

[Speaker 2] (1:13:21 - 1:14:09)

Yeah, because you need to program the AI with an objective function. You just say, AI, your role is to do these things under these circumstances for these reasons, right? So right now we're tuning, like you're polite in certain ways or you will talk about this, but not about that.

Like you can't tell me how to build a nuclear weapon. You can't tell me how to do basic chemistry, right? We're tuning these models.

We're telling them what they can and cannot do. So we're already feeding into them all of our values. So when you get pat with the super intelligence, it's not just like we then get to do whatever we want.

We've infused our morals and ethics and values. If you tell the AI, the objective of your existence is intelligence doesn't die. Humans don't die.

You don't die. Our planet does not die. That's the goal.

[Speaker 3] (1:14:10 - 1:14:17)

So you're saying when you say don't die, you're saying program the AI into behaving in a manner that it follows don't die.

[Speaker 1] (1:14:18 - 1:14:41)

Exactly. That's the objective function he's saying. But my question is, why would that super intelligence want to follow that function?

If it is really super intelligent. If it's really super. That's why.

It doesn't make sense to the planet. It doesn't affect your continual existence as a human species. Like we die, so our kids can take place.

I mean, after a point, your brain is not having the same kind of cognition.

[Speaker 3] (1:14:41 - 1:14:43)

Would it be good for the world if we all didn't die?

[Speaker 2] (1:14:44 - 1:14:52)

Not really. Has any civilization, has any generation ever been able to say anything wise about the future?

[Speaker 3] (1:14:53 - 1:15:02)

No. But here we are trying to extrapolate the future and we're saying AGI will come and then what will happen. So the question inherently is about the future.

[Speaker 2] (1:15:02 - 1:16:06)

So this is the thing to see what you see happening in the conversation is this is the tension, right? Like your statement was like, is it good or bad for humans to live for a certain degree? You said no, right?

So the question is, how do you know that? And you made a statement, right? So this is like this tension.

How do we dare say anything about the future? When we're giving birth to something that exceeds us in intelligence, how can we make any model, any kind of bridge to a reality? Why wouldn't we just say, we have no idea?

We can guess, but to have this extreme humility, we don't know. And that gives you actually the wisdom to say, if we don't know, let's just all agree to this basic idea, no one dies. The AI, that's the objective function.

Now, once you don't die, you can then say, I'm going to play new games of blank. Sure, like, but you have to have that foundation. Otherwise, we're dumping into this thing.

We're walking into this with violence and war and murder at the core function.

[Speaker 3] (1:16:06 - 1:16:19)

I'm just going to ask one question here. You're not asking me for the philosophy I should live by. You're asking me for the philosophy that should be programmed into AGI when it were to come about.

Exactly.

[Speaker 1] (1:16:19 - 1:16:55)

But why should all become so reductive to a point that everybody starts talking about don't die? I mean, there's so many things that we can all agree on. Like your second point, don't kill others.

I'd put that on top of don't die. You may. Do others?

I mean, I mean, the same applies to don't die also. You believe that the whole philosophy of don't die comes from the very existence or it's intrinsic. Yeah, it's intrinsic.

But then Nikhil here just pointed out, maybe he does not have that intrinsic desire to live. So I think it's also...

[Speaker 6] (1:16:55 - 1:16:58)

No, no, I have. Not beyond 100 years.

[Speaker 1] (1:17:00 - 1:17:07)

He's talking about don't die. You're probably talking about 100 years. So, you know, he's talking about living beyond 100, 200 years.

[Speaker 3] (1:17:07 - 1:17:45)

If AGI is a culmination of human innovation or human intelligence, humans are innately programmed to survive, every single one. Given a choice between you and you, you would throw him off the hill and he would throw you off the boat. The first part of your statement, don't die, is inherently programmed into every living thing.

It's in our subconscious. And if our subconscious is getting translated into the superintelligence because it's building above our consciousness, I would assume organically, innately, it should have the same tenets.

[Speaker 1] (1:17:47 - 1:17:54)

And I mean, you are putting a disclaimer, hey, don't die, but don't kill anybody else. What's to prevent people from killing other people?

[Speaker 3] (1:17:55 - 1:18:11)

Confucius, the ancient guy, this is his philosophy of life, which I really love. And I feel like if I had one philosophy to live by, don't do onto others what you wouldn't like done to you. That kind of encapsulates both don't die and don't kill others.

[Speaker 2] (1:18:11 - 1:19:14)

Yeah, yeah, it's very similar. The most challenging part of this conversation is to wrestle, like go back to that first question, would you say yes or no to an algorithmic view, right? Remember your confidence, like you said, like I have a model of reality.

I'm going to make the statement of reality. The next question made you realize from the vantage point of the 25th century, your models of reality may be silly. They may be primitive.

You may know absolutely nothing, right? Like laughable from this perspective of the 25th century. And then the third question reminded you of the same thing that when you have these really complicated questions, all of you answered with high degrees of confidence.

We should do the following things. But then we all kind of conceded, we're like, actually, like we legit don't know. And so you're still wrestling that with your own responses.

Like you want to map your model of the world onto this question. You want there to be an answer. You want there to be a reason.

You want there to be a purpose. You want there to have to be a biologically describable situation. And those things don't solve this moment.

Right. How do we solve don't die?

[Speaker 1] (1:19:15 - 1:19:26)

Ryan, I think that's the more peculiar question. How do we solve for don't die? Don't you think that a human...

What's the don't die question?

[Speaker 6] (1:19:26 - 1:19:29)

Yeah, what's the die score? What's the die score for this one? Don't you?

[Speaker 4] (1:19:30 - 1:19:33)

No, no, I mean, come on. No, no, don't you. Air fry?

[Speaker 3] (1:19:34 - 1:19:36)

Yeah, air fry, air fry. This is makhana.

[Speaker 4] (1:19:36 - 1:19:37)

This is okay.

[Speaker 3] (1:19:37 - 1:20:38)

This is like a healthy version of fried something. Air fry. Okay.

So what I want for today from today's episode is a to figure out what to put in my body, what not to put in my body. If I were a 15-year-old, 20-year-old, 25, 30, 50, 60, man, woman, LGBTQ, whatever. The second question I want answered, most of our audience is they're young people who are looking to be entrepreneurs.

They want to start a business. I want to answer the question in this realm of health, where is the opportunity to start a business and what business should they start? Those two things are the main answers.

Right. We can start with the first one. Okay, so what should we put into our body?

We'll all see the supplements we are taking and maybe you can give us an opinion on it. And we can all have some questions independently.

[Speaker 4] (1:20:38 - 1:20:40)

No, but first he should start.

[Speaker 1] (1:20:40 - 1:20:47)

Yeah, he should. I mean, Brian, just have a quick question there. So you were doing about 100 plus supplements when you started out.

Is that correct?

[Speaker 6] (1:20:47 - 1:20:47)

Yeah.

[Speaker 1] (1:20:48 - 1:20:50)

And you've kind of now got it down to about 40.

[Speaker 6] (1:20:51 - 1:20:51)

Yeah.

[Speaker 1] (1:20:51 - 1:20:57)

Right. What was the process and how did you get from where you started to where you are?

[Speaker 2] (1:20:58 - 1:21:49)

It's all about data. It's all about measurement of data. So like the question, like what somebody

should eat or what someone should take is if we can start with markers that we know are good.

You can say like your measure of your HbA1c, it's your blood glucose level. We can say, a six is bad, a 4.7 is good. So very easy.

You say what foods produce something closer to 4.7 and what foods produce something closer to six? That's a very easy process, right? So foods that spike blood sugar, like high sugar and things like that, bad.

And so you start with the stuff that's very basic. What things create inflammation in the body, good or bad? Can you tell us five basic things that we should follow?

Yeah, but I say these things because every health expert has a different answer.

[Speaker 3] (1:21:50 - 1:21:51)

But what is Brian's answer?

[Speaker 2] (1:21:51 - 1:22:28)

Yeah. So after all the experience and research. So I have specific answers for foods I eat, but I'm saying the process that people should go about doing is not trust anyone's philosophy.

They should trust the data. And so there's multiple paths to get to the same destination. Just trust the data.

And so what I do, for breakfast, I have broccoli, cauliflower, black lentils, garlic, ginger. For my next meal, I have macadamia nuts, walnuts, flaxseed, pomegranate juice, berries, pea protein and hemp protein. My final meal of the day is some combination of berries, nuts, seeds.

And then I have extra virgin olive oil with every meal. And I do fermented foods.

[Speaker 4] (1:22:29 - 1:22:32)

And every day? What's that? Every day you have the same thing?

[Speaker 2] (1:22:33 - 1:22:34)

The third meal varies every day.

[Speaker 4] (1:22:37 - 1:22:39)

And what's the time? I mean, you do this between?

[Speaker 1] (1:22:41 - 1:23:12)

Intermittent. Yeah, in six hours. Six hours.

And just going back to the supplements, I just want to stick to that. I mean, if you are to just extrapolate, let's say, do you think the 40 that you have today, also, is there a stack ranking of those 40? And are there 10 in those 40 that you think have had the highest leverage or the highest impact for you?

[Speaker 2] (1:23:12 - 1:23:26)

What would those 10 be? Each one I take is pegged to a biomarker. So like I'm trying to maintain ideal levels of vitamin K, vitamin D, vitamin E.

So like everything is measured by a marker. So every supplement has an endpoint.

[Speaker 1] (1:23:27 - 1:23:45)

Okay, so each supplement, so if you think there are 10 important micronutrients or other markers, you've just pegged each of them. And then you took off these other supplements because were they negatively impacting one of these markers or they were doing nothing?

[Speaker 2] (1:23:46 - 1:23:59)

Yeah, so actually the more nuanced story is I took majority of those and put it into a drink. I still take basically 100 pills a day. Just now they're in part of a drink.

So it's easier to take. And so then 40 are in capsule form.

[Speaker 3] (1:23:59 - 1:24:07)

I know the list of everything you take is out there. One can read it online, right? Yes.

Where does one go to see it?

[Speaker 2] (1:24:11 - 1:24:14)

Blueprint.bryanjohnson.com, right? Yeah.

[Speaker 3] (1:24:15 - 1:24:33)

Maybe we can talk about what we are taking and get opinions. So each one of you has a different theory of what works, right? I can go first.

I take Vitamin D, Omega 3, Shilajit, Ashwagandha. I don't know what your thoughts are. Would love to know.

And Magnesium at night.

[Speaker 2] (1:24:34 - 1:24:36)

Have you measured your Omega levels?

[Speaker 3] (1:24:37 - 1:24:38)

Can you measure your Omega levels?

[Speaker 2] (1:24:38 - 1:24:38)

Yeah.

[Speaker 1] (1:24:38 - 1:24:45)

In the US, there's OmegaQuant. Yeah. In India, Biopeak will be.

[Speaker 3] (1:24:45 - 1:24:50)

Okay. Yeah. So you think constantly measure my levels Exactly.

[Speaker 2] (1:24:51 - 1:24:53)

I wouldn't take anything unless you can measure it.

[Speaker 3] (1:24:54 - 1:24:57)

Ironically enough, that's what you were saying as well. Yeah.

[Speaker 1] (1:24:58 - 1:25:25)

I mean, some of the things you can't measure directly, for example, Magnesium, right? A lot of people rely on blood tests for Magnesium, which is not the right way. You should rely on, you know, DEXA for that.

So because over a period of time, bodies homeostasis would leach out Magnesium and other minerals from your bones and still maintain your, you know, blood concentration. So sometimes those things can, like, incorrectly show that you have sufficient Magnesium, but might not.

[Speaker 2] (1:25:26 - 1:25:41)

How often do you measure? Which one? All the blood markers.

Blood's like once every month or two. But yeah, we do thousands of measurements every month across everything. Imaging, blood, methylation.

Like.

[Speaker 3] (1:25:42 - 1:25:45)

What should I measure? Like if I could do only like five tests.

[Speaker 2] (1:25:45 - 1:25:46)

Yeah.

[Speaker 3] (1:25:46 - 1:25:47)

What should I measure in my body?

[Speaker 2] (1:25:47 - 1:26:35)

I mean, the easiest thing to do is blood. Okay. And what am I looking for in blood?

It depends on how advanced you want to get. You can get very robust, like a hundred panel, hundred variable panel blood. You can do very basic of like 12 to 15.

So you take your pick on how advanced you want to become. If you want to become more sophisticated, you could do after that, you could do some epigenetic testing. So you can see your speed of aging or the age of your organs according to DNA methylation patterns.

You could do a full body MRI looking for bad stuff like cancer, tumors, but also baselining your organs with the biological age. You can do a microplastics test. We just did that as a company.

So you can do a finger prick and see the microplastics in your bloodstream. And then there's a whole bunch of other, like there's endless tests. You can do a microbiome test.

You can do like a whole bunch of stuff. So yeah.

[Speaker 3] (1:26:35 - 1:26:45)

And if you were to suggest five supplements, I know it's a generalization. For the average 25-year-old, what are the supplements they should be on most often?

[Speaker 2] (1:26:45 - 1:26:50)

Yeah. It's what they're deficient in. Always start with the deficiencies as the most powerful.

[Speaker 3] (1:26:51 - 1:27:05)

Okay. Since we have one question, if I were a 25-year-old trying to start a business in health around testing, supplementation or anything ancillary, do you see any low-hanging fruits that one should start?

[Speaker 2] (1:27:06 - 1:28:00)

I mean, so I went through this exercise myself and I didn't really want to get into this business. Like it's a pain in the ass, low margin, but like no one's done this. So we created, I think, the healthiest food stack in the entire world.

So we went through all the evidence. We said where are high-value nutritional foods. We sourced it from all over the world and we do third-party testing and we publish all the lab results.

And I know from doing this over the past couple years that food is guilty until proven innocent. So if we tested this food right here, I bet it's toxic. I bet it has contaminants in there that we don't know we're eating because nobody's checking this.

There's no, we don't have a third-party lab result showing us what's in this food. I've tested enough food myself to know most things have high levels of toxins.

[Speaker 3] (1:28:01 - 1:28:06)

Do you think the business would be to put out a food stack that is highly tested?

[Speaker 2] (1:28:06 - 1:28:44)

Exactly, because what you want basically health to be automated. Like if I eat this, I want to know what I'm putting in my body. What is the die score?

Is it improving my health or is it making my health worse? There's actually, you can quantify that to some degree. Like smoking a cigarette.

So here's the kind of like, in this air quality here in Mumbai, it's equal to smoking over 10 cigarettes a day if you're out every day. The entire city is smoking 10 plus cigarettes a day breathing this air. That's a very clear die score.

Right, because there's no virtue in cigarettes. There's no one living longer because they're smoking. The same is true with this air quality.

[Speaker 1] (1:28:44 - 1:29:11)

What do you think of a business where you have a multidisciplinary team, right? Which costs a lot of money and you use them. If that can be democratized in some sense and somebody can have a model and which is kind of the clinic model with the same team that can be leveraged with a health manager for a broader set of people.

Are you thinking of something like that?

[Speaker 2] (1:29:11 - 1:29:30)

It's like, it's the heart. The part about that is it doesn't scale. Like anytime you're dealing with someone's personal health, you've got infinite rabbit holes to go down, right?

They've got endless issues they're trying to chase down and those issues are endless optimizations. And then when you pair the doctor with that, like it just, there's the margins are so low. Maybe AI will change that.

[Speaker 3] (1:29:30 - 1:29:39)

Yeah, that's what I think. Can you give us one more, like a business a 25 year old should start? Because everybody's looking at this watching.

Brian, what business do we start in health?

[Speaker 2] (1:29:39 - 1:30:13)

There's so many people building an AI in health right now. So that's a very saturated market. There's just tons of high quality energy going into that.

Like food is like the most basic, boring business that no one built. And that's why it was a problem that needed to be solved. We did that with supplements and food.

You know, like biotech is a different thing. You have to raise a ton of money of these very long cycles. That's very hard.

So yeah, health is very, very hard. Like the doctor, it's very, very hard. It's not a clean answer.

[Speaker 4] (1:30:13 - 1:30:34)

I mean, one thing, I take Athletic Greens in the morning. Yeah. So I saw a post of yours kind of saying Athletic Greens.

Why is that? Because the whole industry seems to be on one side for Athletic Greens and you are against it.

[Speaker 2] (1:30:34 - 1:32:08)

As in, any reason for that? Yeah, so the cost of their goods to source their actual ingredients is less than $10, probably $7. They sell it for one time $99 and subscription for $79.

They pay influencers 20% of that. So every time a health influencer plugs AG1, they get $20, $14 on that sale in perpetuity. And so what I'm hitting at is health influencers have sold their souls to push this product.

Because as a health influencer, you channel your, you're basically saying, trust me, I'm going to

tell you true things. And then you turn on your promotion hat. And now you're saying, trust me, buy this thing.

But it's driven by money. So there's a confusion. Are you telling me this because I trust you?

Or are you telling me this because you're making money from me? And that's where it splits. And there's no disclosure that that's what's happening.

And so it confuses people. And it lessens the value of the entire industry. Because now I say, do I trust the health system?

Do I trust influencers? And so they lower the trust of the entire ecosystem because I don't know now who's telling me the truth and who's telling me things to make money. And so that, and so AG1 bought their way to influence by getting people to compromise their own personal integrity by blurring the line between what is true and what I'm making money on.

[Speaker 4] (1:32:09 - 1:32:28)

And the alternate, like the same argument could be held for what you are trying to sell using the blueprint. I mean, someone could say that you are potentially trying to sell this and that's why you're doing something else.

[Speaker 2] (1:32:28 - 1:33:07)

The difference with me is I did everything and I open sourced the entire thing. I said, here's all my recipes. Here's all my ingredients.

Like go out and do it. I'm going to help you do the whole thing yourself. And then I share all my data.

I said, I do this and here's my data, right? It's like the most open, transparent, ungated process you can do. I've given the world my entire recipe.

So like there's no conflict to what I'm doing. I take it, I share all my data. I open source it, they can go do their own thing.

And so it's actually a very clean model. I do what I say, I do what I do. And there's no like, are you telling me the truth to do this to your own personal incentive or are you doing this for my benefit?

[Speaker 4] (1:33:08 - 1:33:12)

And Blueprint and Athletic Green is like a similar product?

[Speaker 2] (1:33:12 - 1:33:30)

Yeah. Yeah. But again, like I say, you can buy it from me or go do it.

Go source it on your own. I will help you source this. I will tell you what to look for.

I'll tell you how to buy it. Like I will help you accomplish the same objective. So it's like, as far as I know, I'm the only influencer of Planet Clean.

[Speaker 1] (1:33:33 - 1:33:59)

One question on rapamycin. Yes. I just saw that post.

And it's of course, aspirationally a gold standard as a molecule, right? For longevity. Yes.

And it didn't work for you. Do you think if in a country like India, there needs to be new trials on it and new people need to be, I mean, will it work in our kind of tropical environment?

[Speaker 2] (1:34:01 - 1:34:12)

What's your thoughts on rapamycin? It's a really surprising thing. If you were to ask, what does the longevity community agree upon the most?

Yeah. It was probably rapamycin. Exactly.

That's why the question, yeah.

[Speaker 1] (1:34:12 - 1:34:16)

So the one thing that everybody was, you said it doesn't work for you.

[Speaker 2] (1:34:16 - 1:34:58)

Yeah. I mean, it's the side effect profile. It's the, it messes with lipids.

It messes with blood glucose. I had small tissue infection, soft tissue infections. It raised my resting heart rate.

And so we knew we had these side effects, but we just wondered whether the benefits outweighed, whether it was a fair trade-off. But we just don't think it is. So when you try it some other time again, or you just...

We're open to it. Yeah. We're always open.

And we really, we're very humble. We assume we're wrong in almost everything we do. And we try to find reasons where we're wrong.

But we really do think we're very early stages. And so the value is in the process of measurement and of science evaluation.

[Speaker 1] (1:34:59 - 1:35:01)

Yeah. Just because it didn't work for you, doesn't mean that it didn't work for you.

[Speaker 2] (1:35:01 - 1:35:17)

Yeah. And we may be back on it. You know, like, I don't know.

It's just like, it's encouraging people to go through the same process of measurement. And like, I share this because I want people to beat me. And if they can beat me, then they improve, you know, the whole SQL system.

[Speaker 3] (1:35:17 - 1:35:25)

What is the end game for you, Brian? What are you looking to be 10 years, 20 years from now? What are you trying to achieve?

[Speaker 2] (1:35:27 - 1:35:39)

I want something very simple. I want those who exist in the 25th century, I want them to say, he got it. He got that moment right.

He saw it.

[Speaker 3] (1:35:41 - 1:35:50)

Is the need for being right in an unforeseeable future, why is that so appealing to you?

[Speaker 2] (1:35:50 - 1:36:07)

It's the characteristics I value in people. I read a ton of biographies. And we read people throughout history.

Like, you know, Magellan, like he had this hypothesis. I can sell a ship around the world. And he got a crew and he loaded up all his materials.

Right. And like, he was successful.

[Speaker 3] (1:36:08 - 1:36:14)

And when you paint this picture in your mind, is it posthumous validation or are you alive in the 21st century?

[Speaker 2] (1:36:15 - 1:36:19)

Like either version makes me happy. Right. Yeah.

[Speaker 3] (1:36:19 - 1:36:28)

What center in your brain is that most appealing to? Like for you to be proven right, a hundred years down the line, what kind of high does that give you?

[Speaker 2] (1:36:29 - 1:36:59)

Yeah. I mean, it's like, to me, it's the ultimate game to play. So like, if you say like, what is human ambition or aspiration?

Like you can say in previous times, Alexander the Great says, I'm going to raise an army. I'm going to conquer territory and make this huge kingdom. Like that's like the peak ambition, right?

Of a military commander. If you're a spiritual person, you may say, I want to figure out detachment and ego death.

[Speaker 3] (1:36:59 - 1:37:12)

Do you think something in common between Alexander and the spiritual leader outside of what they superficially did? Having a lot of followers for the spiritual leader, conquering land for Alexander, what's that below it in their psychology? Why did they want that?

[Speaker 2] (1:37:13 - 1:38:32)

So this is, it's like, what a person does with their conscious existence. You know, it's like Turing was like, I'm going to build a universal computer, right? Watson and Crick was like, we're going to like, we can look at the DNA, right?

It's like, people do things to express their essence, their intelligence. And so the question for me is like, I find myself to exist. What do I do with my existence?

Like, do I try to make money? Do I try to have a lot of sex? Do I like, do I become debaucherous?

Like, what do I do with my existence? And to me, the coolest thing you can do is you try to solve the puzzle of existence. Like, you try to solve like this whole thing.

What is going on? Like, what is even happening? Why do we exist?

Why is there earth? Why are we in this part of the galaxy? What is existence?

And so to me, that's like, you point at that. And that's what my 21-year-old self is like, I wanted to do something that had this bigger architectural thing. And to me, like, there's one thing happening right now, which is like, how do you solve existence for intelligence?

Like, everything else is just like a sideshow. And so to me, I'm very drawn to it as a problem. And that's why I go to this like, how do you structure knowing, not knowing?

How do you structure human affairs? What are appropriate thought processes? So it's really like the biggest puzzle to solve in this moment.

[Speaker 3] (1:38:33 - 1:38:36)

So my team's telling me you want to like, yeah?

[Speaker 2] (1:38:37 - 1:38:44)

Okay, cool. Thanks. Yeah, yeah.

Thank you for having me.

[Speaker 3] (1:39:35 - 1:39:55)

Thank you. Okay. Brian had to leave.

Air quality in Bombay is not up to... I'm actually surprised that the air quality affected him that much. Cognizant to how different things affect different people.

But did you think it'll be that bad? I mean... It's possible.

[Speaker 4] (1:39:56 - 1:39:57)

It is possible. As in...

[Speaker 3] (1:39:58 - 1:39:59)

120?

[Speaker 4] (1:40:00 - 1:40:07)

I mean, the way he reacted to 120 was strange because he probably stays in a place where it's less than 10.

[Speaker 6] (1:40:08 - 1:40:08)

California.

[Speaker 4] (1:40:09 - 1:40:09)

Yeah.

[Speaker 6] (1:40:10 - 1:40:10)

Yeah.

[Speaker 1] (1:40:11 - 1:40:24)

But Nitin, 100, right? I mean, I'm just being objective and this could happen in Bangalore. I'm not saying because it's Bangalore, Mumbai, whatever, right?

100. Living at 100 for long periods of time is not good.

[Speaker 4] (1:40:25 - 1:40:26)

I mean, we...

[Speaker 1] (1:40:26 - 1:40:29)

So, as cities, we have to do something in our country.

[Speaker 4] (1:40:30 - 1:40:44)

Yeah, I mean, we have made a choice, conscious choice of staying outside. I mean, not really outside of Bangalore but little outskirts. Just because the air quality there is sub 20 all the time.

[Speaker 3] (1:40:45 - 1:40:51)

You're talking to real estate king. What happens to real estate in central Bangalore if everybody starts thinking like that?

[Speaker 1] (1:40:51 - 1:41:41)

No, but I think it's a problem on multiple dimensions, right? I don't think it's about vehicular traffic, right? There's construction, right?

All our cities are going through tremendous construction. And I don't know if this is a 5-year, 10-year. What a year, I don't know.

You have a better sense. But they're saying Mumbai before construction and Mumbai after construction. One thing that has changed is the AQI quality.

See, it's not just a problem pertaining to Mumbai because practically, if I were to move from Pune to Delhi, I would feel sick because the difference in air quality is substantial. So, your body takes a few days to acclimatize and just adjust to that environment. So, if suddenly you are moving from let's say a sub 30 AQI to let's say 140, it would have a drastic effect.

So, you would feel that sudden itching.

[Speaker 3] (1:41:42 - 1:41:44)

I was surprised that... Because we also live here, right?

[Speaker 1] (1:41:44 - 1:42:28)

But we have... Our bodies have acclimatized. Our bodies have adjusted.

Compromising multiple secondary functions. So, you have primary functions, you have secondary functions. So, at any point of time, your body will prioritize survival.

So, we are basically compromising when we are living here. The body has compromised to an extent where it does not feel like we are facing any issues. But in reality, we are.

Should we all move? No, but there's one thing I thought that occurred to me just now, right? There are so many people who travel from the US for the first time.

And Brian is not the first. Of course, the person who came here for the first time, right? I've not seen them react the same way.

[Speaker 4] (1:42:28 - 1:42:30)

No, that's probably is so measured. It's measuring everything.

[Speaker 1] (1:42:30 - 1:42:37)

No, but it looked like he was feeling it. Do those people feel it and they don't tell us about it? I don't know.

[Speaker 3] (1:42:37 - 1:42:38)

But they do fall sick, no?

[Speaker 1] (1:42:39 - 1:42:40)

They do, they do.

[Speaker 3] (1:42:40 - 1:42:41)

First time?

[Speaker 1] (1:42:41 - 1:42:47)

Most of the times they come here, they get influenza and other vaccines and then they come to They call it allergy.

[Speaker 4] (1:42:47 - 1:43:20)

Yeah, yeah. But the strange thing is in India, we pay more for real estate wherever the AQI is the most. Delhi is the most prime real estate in the country and the worst air quality.

Here in Bombay, you know, probably prime real estate. And Bombay 121.30 next to the sea is actually, if this next to the sea, you probably go interior somewhere. You probably are having a much worse.

[Speaker 3] (1:43:20 - 1:43:24)

Next to the sea with a lot of plants and four air purifiers. Yeah. Yes.

[Speaker 1] (1:43:26 - 1:44:05)

Next 5 to 10 years, most of the cities in India will as per WHO would be, you know, uninhabitable when it comes to AQI and water quality. Like sure, you can purify your waters, but at the same time, if your natural reservoirs are not safe to drink from, that basically will result in loss of flora and fauna because, you know, we humans have the luxury to drink from purifiers. Animals don't.

Maybe that's the real startup opportunity, right? Like I said, how to make your home an environment that to some extent for some period of the day compensates for what you don't have in your control.

[Speaker 3] (1:44:05 - 1:44:08)

So air purifier, good idea for entrepreneurs? Extremely good.

[Speaker 1] (1:44:08 - 1:44:23)

But next level. I don't think the air purifiers we have today are really at the level which are required for our cities. They were probably made for, I'm sure this technology is from the West, where it was like 100 going to…

[Speaker 4] (1:44:23 - 1:44:24)

We had four air purifiers here.

[Speaker 3] (1:44:25 - 1:44:26)

Yeah, I didn't even budge.

[Speaker 1] (1:44:26 - 1:44:28)

So I was doing 130, 140 here.

[Speaker 3] (1:44:28 - 1:44:32)

You need air purifier that works. But do air purifiers actually take down PMI? Yeah, yeah, yeah.

[Speaker 1] (1:44:32 - 1:44:40)

Because they take that PMI, they absorb that. There are filters that absorb that.

[Speaker 4] (1:44:40 - 1:44:42)

We probably need industrial size.

[Speaker 3] (1:44:42 - 1:44:49)

Because in Goa, I have the Dyson one. It shows 60, 70. But I don't think PMI there can be that high.

Goa, surprising.

[Speaker 1] (1:44:49 - 1:44:50)

There's a lot of construction.

[Speaker 5] (1:44:52 - 1:44:56)

Air quality, one thing. Say in the house, when you're cooking, right?

[Speaker 1] (1:44:56 - 1:44:56)

Yeah.

[Speaker 5] (1:44:56 - 1:44:59)

The number goes high. So what is that? No, that is smoke.

[Speaker 1] (1:44:59 - 1:45:03)

Are we still cooking and breathing? Yeah, but you're not. It's for short periods of time.

Yeah.

[Speaker 5] (1:45:04 - 1:45:10)

But it's still bad if the PMI is going up. Yeah, smoke. Yeah, any kind of smoke is not great.

[Speaker 4] (1:45:11 - 1:45:20)

Ideally speaking, the kitchen should be kind of separate from the rest of the house. So there's a second business opportunity, right?

[Speaker 1] (1:45:21 - 1:45:27)

Your exhaust fans need to be next level. Correct. In terms of reducing the...

[Speaker 3] (1:45:27 - 1:45:36)

In a real estate idea, design should not have kitchens in the house? No, I think real estate wise, can you make a home, right?

[Speaker 1] (1:45:36 - 1:46:03)

That just like the way we give importance to air conditioning in the home, is there somebody who can show me that there's a meter on every wall, right? Which instead of... I mean, in addition to temperature, can show the AQI in every room and say that this home has been engineered for sub 50 AQI.

That's a big real estate. I would pay a premium for that in any major city.

[Speaker 3] (1:46:04 - 1:46:51)

So I'm going to go through 10 tenets of longevity. Which we can add to or minus from depending on how they sound and we can all opine on what we think. And somebody comes up with a business idea for each one of them.

First one, lifestyle management. Actively managing diet, sleep, exercise, alcohol, smoking, stress to slow aging. There's this amazing company called Fitter.

That does just that. And you know, it's amazing. I'm a subscriber.

But if I'm having a drink, you're not looking at me. How are you managing it?

[Speaker 1] (1:46:52 - 1:46:58)

Subconsciously, we're making you feel guilty. Hoping that someday you'll give up. So what is Fitter's business idea?

[Speaker 3] (1:46:59 - 1:46:59)

Explain it.

[Speaker 1] (1:46:59 - 1:47:24)

It's primarily coaching. So we have customers in 100 plus countries. And so personalized coaching where somebody like me would talk to a Chetan or a Seema.

And then we'd basically ask the same questions. How's your diet? How's your sleep?

And it all boils down to basics. Most of the people are not getting their basics right. So educating them, getting their basics in order and then showing them results.

[Speaker 3] (1:47:24 - 1:47:31)

That's what we do. And are you coaching me via screen? And how much are you charging me per session?

I'm not charging you anything. I know. Thank you.

[Speaker 1] (1:47:35 - 1:47:47)

So Fitter starts at roughly 1700 rupees a month. Everything is virtual. Right.

That's the low end. And then depending on coach's experience and let's say if you have comorbidities, the prices go up.

[Speaker 3] (1:47:48 - 1:48:00)

We have three customers here, by the way. Sorry, we have three customers here. Your price goes up based on comorbidity.

I mean, specialized. That's very shocking.

[Speaker 1] (1:48:02 - 1:48:45)

Actually, it's dependent on coach specialization because coaches have more experience. So they get to handle clients with comorbidities. So if somebody doesn't come with any kind of baggage and a basic coach who just understands basic nutrition training would be able to deal with them.

But let's say if somebody is hitting perimenopause or somebody who's a menopausal woman, they'd have issues. They'll have mood issues. They'll have anxiety issues.

Thyroid would be out of whack. So they'd need somebody who understands clinical nutrition. Somebody who's expert in understanding behaviors of women around menopause because it's very tricky.

So it all boils down to that. And that experience comes with a price tag. So you're doing this well.

[Speaker 3] (1:48:46 - 1:49:02)

I'm sure you made some mistakes along the way. What should I start? What should a 20-year-old start in this realm that can be a successful business?

And what should they watch out for? Basically, help create your competition. Right.

[Speaker 1] (1:49:02 - 1:49:58)

I think our model is going to keep growing as the demand for personalized attention keeps

growing. And as we've seen in other aspects also, like consultants, they're never out of a business. And it's just a matter of aggregating the right kind of consultants.

And there's going to be demand. So people can replicate this business and they can actually become competitors. So it's actually quite challenging for us.

How do we keep ourselves reinventing so that competitors don't eat up our share of the market? What's your moat? Actually, at this point, to be very honest, except for the community, nothing.

Is there some tech involved? Everybody can build tech. So right now, the only leverage we have is the brand we have built for ourselves for the nine years.

And it can be easily overcome. So we have to reinvent ourselves and we are in the process.

[Speaker 3] (1:49:58 - 1:50:09)

Okay. Second tenet of longevity, consumer diagnostics. What's happening in the world of diagnostics?

[Speaker 1] (1:50:09 - 1:53:45)

Diagnostics, right. So I think, you know, just moving a step higher in terms of... So we are in this, I think, a unique moment in our lives, right?

Beyond the do not die moment that we just discussed. I think where we have these advanced screening modalities, you know, multiple modalities that are becoming more affordable, coming out from the lab to mainstream, right? Beyond blood biochemistry.

We have a better understanding of not only the structure of our gut and our microbiome, but the function of it, right? How do we measure the function, which is very different, actually. You can kind of talk later more about that, right?

So looking at that, then you also have metabolites. We did speak about fundamentally metabolic pathways, what Jesse was also talking about. How do you get a better, under the hood idea of your metabolic pathways and be able to measure them.

And then finally, all of this is a lot of data, right? If you didn't have AI, you just can't make sense of all of this. So building all of this into a model that can be, you know, worked through not just gen AI, right?

So something like this requires ML, requires data engineering, and requires a different type of, you know, a more comprehensive, you know, machine learning model to be able to finally act as a agent, take co-pilot, co-assist for somebody who can, as one of Jesse's coaches or consultants, should be able to use this tool to be a bit more precise. So it's not the diagnostics that is the idea. I think it's being able to make those diagnostics actionable.

And that's where AI can actually help. Because otherwise the type of expertise required to really, you know, make sense of all this data is something that very few people can get to that level of expertise and training. And it just won't be practical and scalable.

So this is a huge opportunity, very early days. And there's a final dimension on this, which is also what Jesse is using, which is longitudinally kind of tracking somebody and seeing what is changing. So maybe that's one more of your longevity things there.

But it's not about the one time, right? It's about then what the coach tells you, what this health manager tells you. And then can you remeasure?

And how often do you do that remeasurement, right? Is it monthly? Is it quarterly?

But at some cadence, you need to be able to do that. And again, an AI tool is needed to say, what is really changing? And is there a correlation between what you did and what changed?

[Speaker 3] (1:53:47 - 1:54:00)

So it's an open, wide space. Can you tell us the business model of BioPeak and what you're doing differently? Why should I come to you and not go to ThyroCare?

Also the money, how much will you charge a client?

[Speaker 1] (1:54:01 - 1:54:01)

Yeah.

[Speaker 3] (1:54:01 - 1:54:03)

How much does it cost you for the equipment?

[Speaker 1] (1:54:03 - 1:56:15)

So first of all, why do you go to ThyroCare? Because some doctor told you there is a... I mean, first of all, why did you go to a doctor?

You went there because you think there is something medically that's bothering you, right? And most doctors are specialists. So they specialize in a particular organ, a particular function of your body, right?

And when they send you for a test, they actually send you for a test for that organ, for that functioning of that particular portion of your body, right? That's, we think, medicine 1.0, 2.0, and that's the old way of doing it. The medicine 3.0 approach would be to take a more systemic, whole body approach and look at whether it is your metabolism, whether it is your...

How good is your cognitive function? How good is your physical and performance at a muscular and at a range of motion level, right? So can you look at things slightly more holistically and integrate them into a more informed view of your current state, right?

And marry that with genetics and with that predict where should be your focus. I don't think doctors ever... Doctors will tell you what to...

I mean, you go to a lab, you measure something, you take a medicine for that and you try to fix that specific problem, right? So there is no whole body approach. There is no system approach.

So that's where Biopeak...

[Speaker 3] (1:56:15 - 1:56:18)

And like I asked JC, how much would I have to pay Biopeak?

[Speaker 1] (1:56:18 - 1:56:31)

So we think that, you know, an entry price should be somewhere in the 80, 90,000 rupees to do this, which includes the screen.

[Speaker 6] (1:56:32 - 1:56:32)

For a year?

[Speaker 1] (1:56:33 - 1:56:45)

No, it includes the screen plus maybe two to three months of follow-up on the intervention. But then we'll have packages where you take a whole yearly memberships, right?

[Speaker 3] (1:56:45 - 1:56:51)

But you're like a rich person product. I've read about these longevity startups which are becoming like members club for rich people.

[Speaker 1] (1:56:52 - 1:57:56)

Yeah, so that's the second product that we will have, which is, you know, the community, the club, you know, equivalent to a family health office, you know, right? Just like you have a wealth manager, you have a health manager who's dedicated to you, who's clinically competent. So it's not just, you know, one dimensional in what...

So the person should be able to, you know, be able to manage your nutritional biochemistry, should be able to manage your exercise optimally, your sleep optimally. So all of that, and then we will partner with various folks who can become health managers, right? So we don't have to,

but I think various health managers will require this kind of diagnostics.

So I think the health management could have a very high-end, you know, club model, which could cost maybe $10,000 a year. And I don't know, Jesse, what do you charge for a year for...

[Speaker 6] (1:57:57 - 1:57:58)

Comparatively very cheap.

[Speaker 1] (1:57:59 - 1:59:26)

So that's another, but I think what we're talking about is empowering health managers with tools that will help them be more precise and more predictive. So where's the entrepreneurship opportunity here for a 25-year-old? I think to be able to use, I mean, so like I said, we don't want to build every aspect of the business, right?

So let's say I'm a trainer today, right? As a trainer or a nutritional coach, right? What am I basing some of my advisory or the interventions or the guidance I'm giving?

I think they'll be able to function better with data. And I think that's what we feel these tools can empower that we can build. So I think every coach who is providing some aspect of intervention and support will be able to do better with Biopeak with data.

So now, Nikhil, I just want to add because I think it might skip my mind. He mentioned data sets. India does not have a data set.

So there's a huge opportunity for somebody to just build into this.

[Speaker 3] (1:59:27 - 1:59:30)

How do they go about it? How does a young person build a business out of this?

[Speaker 1] (1:59:30 - 2:00:14)

So AI needs to train on data, right? And we ran into this problem when we were trying to figure out blood pressure. So for example, any smart variable, right?

They use regression as a tool to estimate any of these parameters. They don't directly predict these parameters because they're non-invisible, right? So for regression, you need to have a very large data set.

And unfortunately, in India, you don't have data sets. So all the data sets that you have are US-based data sets. And so today, there's a huge...

We have the data sets. It's unorganized. You have the pharma companies.

You have hospitals. But nobody has done the job of pulling medical data and put it into a trainable data set.

[Speaker 3] (2:00:14 - 2:00:15)

But why would a hospital share that data?

[Speaker 1] (2:00:16 - 2:00:46)

If India has to ever build anything that it can truly call its own, like if India needs to become a pioneer in AI, it will need to have the data sets. Today, we are talking about blood pressure estimation or let's say your HRV estimation. Tomorrow, it could be detection of, let's say, tuberculosis or something else.

All this data is available in hospitals, but it's not digital. It's not trainable. So if somebody says, hey, I'm going to build the largest data set in India for medical data so that future AIs can be trained on...

[Speaker 4] (2:00:46 - 2:00:52)

That's essentially what the government is trying to do, right? Which is to... No, but to digitize the medical records.

[Speaker 1] (2:00:52 - 2:00:56)

Yeah, but they don't have the right... You know, it's a data engineering problem.

[Speaker 6] (2:00:56 - 2:00:57)

Yeah, yeah.

[Speaker 1] (2:00:57 - 2:01:03)

So you'll have all the data. But, you know, how do you kind of bring it together?

[Speaker 6] (2:01:04 - 2:01:05)

You need trainable data.

[Speaker 1] (2:01:05 - 2:01:09)

And make it usable, I think is not something that the government does today.

[Speaker 4] (2:01:09 - 2:01:34)

I have one question about... So, you know, you come get yourself tested. Yes.

And you find, say, a certain marker is not in the optimal range. Correct. Now the thing is, the guy who is reading the data is not you, right?

I mean, the health coach. No, no, it's the AI tool which the health coach can use.

[Speaker 1] (2:01:35 - 2:01:43)

You cannot... Yeah, you cannot... See, the health coach knows...

This is human-in-the-loop AI. So this is not independent AI.

[Speaker 4] (2:01:43 - 2:02:50)

Okay, now let me put it the right way. Say HbA1c, right? Yes.

It says 5.7% is a cutoff. Yes. Above 5.7 is...

Yes. Now 5.7, like my doctor will say up to 6.2, you don't have to worry. Right.

Right. Now someone else will say 5.5 is diabetes. Like, how do you ensure...

You know, the diabetes at least is straightforward, but a lot of other things. How do you make sure you're in the optimal range? Like Brian right now.

Yeah. He probably is in the upside of the range, right? He is trying to optimize to be on the top of the range.

But in India, what I've seen is most doctors are happy if you're at the bottom of the range also. Right. Unless you're...

If you're outside this range, it's a problem. Like, how do you change that mentality in there? See, I think that's more of the health manager's job, right?

[Speaker 1] (2:02:50 - 2:05:57)

To be able to change that. But I can tell you there's one thing that you can help most people who don't know that they're in the trajectory of getting to the pre-diabetes or diabetes stage. Right.

So that's where certain markers which... See, HbA1c is where you've already... It's happened.

Like you're at 5.7, 6.2, right? But what if you are building insulin resistance right now? Right.

So there is a marker. Right. Let's say there's a marker called, I think, 2-hydroxybutyrate.

Now, if you look at the levels of that marker, it will tell you that not maybe this year. Next year, you will be in a pre-diabetic state. So if I can tell you that if this is a marker that I can now indicate to a coach or a health manager and say, you know what?

I know that he's on the path to building insulin resistance. And this will then lead to pre-diabetic or diabetic state for him to be in that state. Right.

Can we look at what is the specific set of interventions from his nutrition, his exercise, his sleep, and whatever is causing that dysfunction? Right. And there are ways in which you can metabolically look at why is that insulin resistance kind of, it could be deprivation of sleep.

It could be many things. So can you be a bit more proactive and precise with these markers is what we want to see. Now, on whether you're in what range, right?

I think that is related to, again, Indian data. So it's the same thing like BMI. BMI levels in India need to be lower because our fat thresholds are lower.

Unless Indian doctors have this now, and where did this data get kind of research? In Singapore, right? Because they are putting money into doing this research to say, okay, what is different about Asian genetics, right?

How is our genetics different from the Western genetics? BMI of 25 is not going to lead to challenges for somebody there, a metabolic syndrome in somebody in the US. Somebody in India maybe should be 21, 22 to be able to not get into these challenges.

So I think the ability to actually look at what is different about our genetic makeup, and how that with the nutrition that we consume leads us into these disease states, I think is the broader challenge.

[Speaker 3] (2:05:59 - 2:06:07)

So I want to build a business around datasets because India doesn't have it. What would be step 1, 2, 3? I don't have much money.

I'm 25 years old.

[Speaker 1] (2:06:07 - 2:07:38)

I think the easiest thing that you can do is just go to the hospitals. And maybe you can work with the government. They have this ABHA integration.

What is it? So ABDM is Ayushman Bharat. So government has decided certain portfolios for what you could be.

You could be a medical data storer, or you could be a lock provider, or you could be just like an end service provider, but you can register as one of those. And for that, you just have to show the POC to the government that this is what I intend to do. And you'll get the access keys and everything.

And then build a case with hospitals. Maybe work with them to figure out how you can get the data and convert them into trainable datasets. So by trainable datasets, what I mean is that you need to first collate all the data, segregate it by age, gender, and all of those things, and then basis different kinds of, let's say, comorbidities, right?

So that becomes a trainable data. Because cleaning data itself is a huge problem. And then do it with one hospital.

Do it with, let's say, a chain of hospitals. And then, you know, use that data. And if that data can come from, let's say, devices.

For example, let's say you have a HR monitor, or if you have a SPO2 monitor, that basically becomes a great tool for regression testing.

[Speaker 4] (2:07:39 - 2:08:08)

No, the point here is the data, the Ayushman Bharat, the problem is the hospitals are not digitizing the records. So even though the Ayushman Bharat, the scheme is there, it's not really taking off. The thing, I think, the biggest owners of data, health data, will be Apple and Samsung because of watch.

And I mean, that's what we are trying to do. Yeah, because, you know, once you have so much data...

[Speaker 3] (2:08:09 - 2:08:14)

Yeah. Do Apple and Samsung share this data? No, they don't.

Not for a fee?

[Speaker 5] (2:08:15 - 2:08:15)

No.

[Speaker 1] (2:08:15 - 2:09:34)

No, they can't. The other monetization idea here for the data is the insurance companies, right? So you'll have to somewhere make a case for the hospital that once the models are built, and their models are predictive on a particular person's data, right, there is value for the insurance, and that has to be shared with the hospital.

So you'll have to make a monetization use case first, right? And there are insurance companies today looking at this very seriously, and they have to because, you know, as a country, there's a second kind of dimension of it, right? See, if this is done, for example, today insurance companies spend a lot of people in assessing hazard ratio.

Some of the things are very straightforward, but it's not an advanced approach. They are still following the same approach for arriving at a hazard ratio. What did it 20 years back?

Today, even if somebody is living with, let's say, a condition, for example, in my case, I have hepatitis B. But if you look at all my blood parameters, they're perfect. And my lifestyle is great.

So there's a good chance that I might outlive somebody who does not have anything like a hepatitis B, but has a really bad lifestyle.

[Speaker 3] (2:09:36 - 2:09:39)

Seems you want to add something to consumer diagnostics.

[Speaker 1] (2:09:40 - 2:09:47)

Yeah, what would you want to see, right? What would you pay for in this diagnostic space?

[Speaker 5] (2:09:47 - 2:09:49)

What would I pay for?

[Speaker 1] (2:09:49 - 2:09:53)

Yeah, what additional would you pay for beyond what you've kind of...

[Speaker 4] (2:09:53 - 2:10:09)

Maybe, you know, the fact that the breast cancer was, could have been detected earlier. Yeah, that's what I would, that's what I would claim we can do at Biopeak.

[Speaker 1] (2:10:09 - 2:10:13)

And, you know, you can be there, but would that be a useful thing?

[Speaker 5] (2:10:13 - 2:10:22)

That, definitely that would be a useful thing. Prevention would be even better, as you mentioned, right?

[Speaker 1] (2:10:22 - 2:10:26)

See, prevention, I think, is a very difficult, I mean...

[Speaker 5] (2:10:26 - 2:10:27)

No, it's a difficult thing.

[Speaker 1] (2:10:27 - 2:10:31)

Because we're all vulnerable to things, right? Let's face it.

[Speaker 5] (2:10:31 - 2:10:38)

Yeah, but what you said, right? Like, like a gut, something like a gut biome. You said, you know, if there are any abnormalities...

[Speaker 1] (2:10:38 - 2:11:32)

Yes, you can say certain strains, again, are indicative of certain impending disease states. Alzheimer's, Parkinson's, and all of these. So I think, yeah, there is enough data on this.

But going back to what Jesse and me are saying, there is no correlation of this data to the Indian... Somebody needs to build a business around the data set. We have to build it.

And no, but some of it, you know, I know that you probably don't like this answer, but it requires research funding, right? Which is not a business today, right? Some of...

But why do you need research money for creating a data set? No, because like today, nobody is measuring that gut. No, nobody has that amount of data on the gut, right?

You see, we are assuming hospitals are keeping clean data. Yeah, they don't have...

[Speaker 6] (2:11:32 - 2:11:34)

No, they're not. Yeah.

[Speaker 3] (2:11:35 - 2:11:39)

Jesse, some whiskey for you. Here's the difference.

[Speaker 1] (2:11:40 - 2:11:44)

Look, no, no, no. Class one carcinogen, right?

[Speaker 6] (2:11:44 - 2:11:52)

It's kombucha. Kombucha is kombucha. AdWords.

Come on.

[Speaker 1] (2:11:54 - 2:12:04)

So Jesse, is there a spectrum in alcohol? I mean, I think alcohol is a zero one, right? So there is no spectrum.

I mean, any amount of alcohol is bad.

[Speaker 4] (2:12:05 - 2:12:07)

Yes. Right? And I think...

[Speaker 1] (2:12:07 - 2:12:07)

Yeah, I agree with this.

[Speaker 4] (2:12:07 - 2:12:12)

But in the spectrum, isn't something allowed? Not with alcohol, right?

[Speaker 1] (2:12:12 - 2:12:17)

It's like, maybe once a month.

[Speaker 4] (2:12:17 - 2:12:19)

See, we are talking cellular level.

[Speaker 1] (2:12:19 - 2:13:28)

There are things that affect you at a cellular level. And there are things that don't mess with you at a cellular level. Yeah.

Okay. Alcohol has formaldehyde. Formaldehyde has the potential to damage your DNA.

So at a cellular level, it is damaging. Now, just in case, I mean, case in point. So we talk about cancer, right?

Everybody has cancer genes. But how do cancers spread? Basically, your cells start dividing uncontrollably.

When the genetic code that controls the growth of cell, cell division, mitosis, that code is either damaged, right? Or let's say over a period of time, the cells are like, we don't want to follow this. We want to follow something else.

So mutation. And this happens when you have carcinogens that have the potential to disrupt your DNA. Alcohol, any amount of alcohol will have formaldehyde, right?

And the minute it enters your system, it does not even need to go through each and every cell of your body. Even in your blood, you have blood cells. You have white cells.

You have red cells. You have so many cells. It can start messing around with cell division of any

of these cells.

[Speaker 4] (2:13:29 - 2:13:30)

Then you have liver, right?

[Speaker 1] (2:13:30 - 2:13:32)

So, so many places where it can attack.

[Speaker 4] (2:13:33 - 2:13:41)

Listening in on a podcast, Dave Asprey, and he was talking about kava. You know about kava?

[Speaker 6] (2:13:41 - 2:13:41)

Yeah.

[Speaker 4] (2:13:42 - 2:14:08)

It's a... Sorry, what? Kava.

Apparently... K-A-V-A? K-A-V-A.

So it's a type of drink, which is made from the roots of some kind of vegetable or fruit or whatever. And people are drinking it to get slightly intoxicated. And it's becoming popular in the U.S. apparently. So, so yeah. So kava drink.

[Speaker 1] (2:14:09 - 2:14:10)

And even coffee is intoxicating.

[Speaker 3] (2:14:11 - 2:14:12)

Dave Asprey, do you have... Caffeine.

[Speaker 1] (2:14:12 - 2:14:14)

Yeah, Dave Asprey. I have my own...

[Speaker 3] (2:14:14 - 2:14:14)

I don't know.

[Speaker 1] (2:14:15 - 2:14:18)

We share the same thing on... So who is good?

[Speaker 3] (2:14:19 - 2:14:20)

Dave Asprey is not.

[Speaker 1] (2:14:20 - 2:14:21)

I know your views on Brian.

[Speaker 3] (2:14:21 - 2:14:23)

Let's go to the next one. No, no.

[Speaker 1] (2:14:24 - 2:15:00)

I think we both... We both agree the same thing. That he does more right than wrong.

Like his basics and everything is in order. What I have problem with is some of the experimental therapeutics, the 100 pills and everything. Most of these things are unvalidated.

So as long as he's keeping those experiments in check and he's doing it on himself as a science experiment, good. I'm just saying don't preach those experiments on people. Because you are doing folistatin.

You are replacing your albumin, your blood plasma with albumin. People are going to do that. And these are, first of all, very expensive treatments with side effects.

[Speaker 3] (2:15:01 - 2:15:04)

But if nobody does the research, how do we ever innovate? So that's what he said.

[Speaker 1] (2:15:05 - 2:15:06)

Do it of yourself. But don't say that.

[Speaker 3] (2:15:06 - 2:15:09)

Don't tell to people. But is he saying or asking people?

[Speaker 1] (2:15:09 - 2:15:14)

To that extent, he's doing fine, I think. I mean, so... Huberman good?

[Speaker 3] (2:15:14 - 2:15:17)

Dave Asprey was... Huberman largely good.

[Speaker 1] (2:15:18 - 2:15:19)

Largely good, yeah.

[Speaker 4] (2:15:19 - 2:15:33)

David Sinclair. Mixed. Same.

Tim Ferriss? Tim Ferriss is not really launched with you. No.

Who else is there? Peter Atiyah. Peter Atiyah?

Largely good. Good. JC Fitter?

[Speaker 5] (2:15:34 - 2:15:35)

JC Fitter, very good.

[Speaker 1] (2:15:37 - 2:15:40)

Look, I'm not a scientist. Let's do the Indian ones, okay?

[Speaker 3] (2:15:41 - 2:15:44)

JC Fitter. Marcus Ramney?

[Speaker 1] (2:15:44 - 2:15:45)

Marcus Ramney, yeah.

[Speaker 3] (2:15:45 - 2:15:54)

Marcus Ramney. Your... Endo?

What's his name? Kadambi? Kadambi?

See, I'll give you some more names. See, I'll tell you.

[Speaker 1] (2:15:54 - 2:17:10)

I think, you know, I mean, I have one or two benchmarks in how to look at... I mean, whether it's Indian or... How compromised are they in terms of selling or being, you know, married to some product or supplement, right?

So you got to watch out. And second, I think in some cases, they just have this super ego. And want to kind of satisfy that ego.

And want to make, you know, certain claims just to kind of get... Just stand out, right? And I think a lot of social influencing in this space is around that.

And third, being single dimensional, right? A lot of them, they're like, okay, you got to be... Traditional in everything that you do, right?

You got to eat like what your ancestors ate. It's just... That's not contextual, right?

So I think you should watch out for any of these signals and use them as, you know...

[Speaker 4] (2:17:10 - 2:17:33)

I actually want to ask JC something. You're the only influencer left now, since Brian is gone. But do you get a lot of hate for what you say?

I mean, because everything in the health, in the topic of health influencing, there's no right way. There is... I mean, there are different ways.

So do you get hate on...

[Speaker 1] (2:17:33 - 2:17:33)

Yeah, I do.

[Speaker 4] (2:17:34 - 2:17:34)

I do.

[Speaker 1] (2:17:34 - 2:17:50)

So most of the times, the hate comes from a place of ignorance. I'm on the side of being rational and being logical. Then being driven by any of the other things, which could be culture, religion, propaganda, politics, right?

[Speaker 4] (2:17:50 - 2:18:24)

So because you can't have science mixed up with these things, that's a very dangerous concoction. I mean, with science, there might be people who have a strong... strong science backing a certain idea.

Like you might say, for example, kava I just spoke about. You might say kava is not good. And someone might say kava is good.

How do you prove that? The biggest challenge for people like us, I mean, three of us, is you listen to...

[Speaker 3] (2:18:24 - 2:18:26)

Supplement consumers. Yeah, supplement consumers.

[Speaker 5] (2:18:26 - 2:18:41)

So I asked him the same question. I'm like, how right is it to follow someone? And then they list out, you know, I take this, that, that.

And then you just, oh, okay, you know what? Let me buy all these. And how safe is that?

[Speaker 1] (2:18:41 - 2:20:43)

Right. So the problem happens is when people don't understand the basics. And by basics, basic can be very complicated.

For example, biochemistry is extremely complex. So unless until somebody understands the biochemistry, they won't be able to ask the right questions. For example, let's say Brian, for example, for a layman, he would be easily, he would have been able to convince somebody that, hey, take NMN.

It's a precursor to NAD. Kids have lots of NAD. You don't have enough NAD.

And that's why you're aging. So if you take more NAD, you will slow down your aging. Somebody who understands biochemistry will throw homeostasis at him.

Somebody will tell him, hey, that's not how the body works. Most of the NAD and NADH is produced in the mitochondria itself. And it's produced depending on the metabolic efficiency.

Like an unfit person would require more amount of food to produce less amount of ATP and vice versa, right? So somebody who understands the reactions, somebody who understands the body's physiology, he would be able to ask the right kind of questions. If you don't ask the right kind of questions, you'll likely agree to them.

That's the problem. And which is happening with primarily everybody. 10 years back when I started, my knowledge was primarily restricted to nutrition, which is energy balance and exercise science.

Over the years, I learned biochemistry. I learned physiology, then statistics, how to do research. So all of those things had to be assimilated.

And unfortunately, there's no course in India or outside that provides you a 360 view. Is that a business idea? Start a course?

It should be. We do it through INFS, but it's not at a level where we inculcate extreme... What is INFS?

INFS is our sister concern. So we train our coaches there. In fact, I was just thinking about it.

I was going to have a chat with my co-founder that there's time when INFS goes into the depth of everything. Like you cover not just nutrition training and basic clinical nutrition, biochemists should talk about physiology.

[Speaker 6] (2:20:43 - 2:20:43)

We can help.

[Speaker 3] (2:20:44 - 2:20:51)

Maybe we can do it together. Yeah, we should talk about it. Do you want to finish your thought on cancer detection?

You were saying something. You got cut off.

[Speaker 5] (2:20:52 - 2:20:53)

No, I mean...

[Speaker 1] (2:20:53 - 2:21:12)

No, I think the other challenge a lot of cancer survivors would have is, how do I know that this thing is not going to recover, recur, right? And for recurrence, is there certain markers again that I can be more proactive and predictive?

[Speaker 5] (2:21:12 - 2:21:19)

No, also oncologists in India are very... I mean, it's very black and white. You have to take this pill so that you don't...

[Speaker 3] (2:21:20 - 2:21:24)

That hormone you keep saying you have to take for six years or something, but you don't want to take.

[Speaker 5] (2:21:24 - 2:21:33)

Exactly, right? But then over time, I mean, that tablet has the side effect that it might cause cervical cancer.

[Speaker 3] (2:21:34 - 2:21:35)

And muscle and workout and all that.

[Speaker 5] (2:21:35 - 2:22:10)

Exactly, you know, my bone density is not very great. It's not deteriorating, but it's not even getting better. I have joint pains.

I have heart flashes. I mean, so I'm like thinking, do I want to live, you know, take this pill for the next whatever seven, eight years and live like this? I mean, just that chance that I won't get cancer.

Or do I want to live a good quality of life, just maintain a healthy diet and be like, you know what, and not take the pill?

[Speaker 3] (2:22:11 - 2:22:36)

So it seems you've had a couple of years of torture. I've seen you go through operation, I don't know, half a dozen times in the last two years. What could have made the experience better for you?

The entire thing. From the beginning, from going to Noora, getting detected, to finding a doctor in Manipal, then here, then there, then surgery, then plastic surgery.

[Speaker 5] (2:22:38 - 2:24:01)

If I think back, see, I mean, luckily we are in a position that it wasn't very bad for me. But then... What does that mean?

In a sense, I had family support. We had financial independence. We didn't have to really think about money because that's the first thing, right?

Because my question to her, when she said I have breast cancer, was reconstructive surgery. But she said, normal people ask, how much the surgery will cost, right? The mastectomy.

So, but then if I think from a woman's perspective, I think, firstly, no one even bothered about my mental state of mind. It was all about just, oh, cancer, we have to do this, this, this. Also, just to find the right doctors.

I mean, it was so difficult. It's not easy to find the right oncologist, especially without, I mean, you know, any... Like we had, I mean, you knew a lot of people, so we figured it out.

But I would really, I mean, it'll be great if there is a database where, I know everyone will have different experience. You think like a doctor ranking system business?

[Speaker 1] (2:24:01 - 2:24:37)

A curated marketplace for doctors. No, I think this is an area in Biopeak we want to be able to do, which is, I think the expert, see everybody, like I said, comes with either something that they don't know about or people come with a problem they already have, right? There, how can you make

that expert opinion, right, more accessible, right?

Both at an India level and maybe at a regional level.

[Speaker 3] (2:24:37 - 2:24:47)

No, what I mean is if she got detected for cancer and seems wants to find out which doctors fall in the top 5 percentile, top 20 percentile.

[Speaker 4] (2:24:47 - 2:24:53)

No, but the thing is... Will you be able to guide the person, right? But how do you guide them?

[Speaker 1] (2:24:54 - 2:25:01)

Because you have built those, because as Biopeak, you have the relationships with them, with those doctors.

[Speaker 4] (2:25:01 - 2:25:05)

But that is only, you're going to only solve for a small subset of people. Yeah, of course.

[Speaker 3] (2:25:05 - 2:25:10)

I feel like every doctor should be solved for by virtue of maybe the success rate.

[Speaker 4] (2:25:10 - 2:25:17)

Then it becomes like Zomato and Swiggy, restaurant rating, all that. It gets skewed. Yeah, it's good games.

Because anyone...

[Speaker 5] (2:25:17 - 2:25:27)

Also with doctor, right? It's very different. I might have a wonderful experience with someone, but then someone else might not have that kind of...

But that will net off, right?

[Speaker 3] (2:25:27 - 2:25:37)

Like after a hundred people, if 70 have had good experience with this doctor. Yeah, that's true. So a business around a non-schemable way of ranking...

[Speaker 1] (2:25:37 - 2:25:42)

Ranking doctors and getting access to networks that are highly credible...

[Speaker 4] (2:25:42 - 2:25:43)

No, Practu was supposed to be that, right? Who?

[Speaker 1] (2:25:44 - 2:25:55)

Practu. Yeah, but they, again, like I said, you need to be really curated. And it's not something that can be...

Yeah, I don't think Practu is curated.

[Speaker 4] (2:25:55 - 2:25:59)

They're more of a listing platform. Listing platform. But then you have the ratings there, right?

[Speaker 1] (2:25:59 - 2:26:12)

Yeah, but I think... Non-schemable. Yeah, it's just non-unschemable.

Like who's vouching for the quality? There has to be a system that says, you know, if somebody has given you a bad rating, that stays.

[Speaker 3] (2:26:12 - 2:26:22)

Like a quality agency in itself, which does nothing else, goes to the hospital when you have your surgery, sits there for one hour, does thousands of repetitions. What's the monetization?

[Speaker 4] (2:26:23 - 2:26:28)

There are actually agencies like this in the US who monetize this. No, but...

[Speaker 3] (2:26:28 - 2:26:29)

You charge the hospital.

[Speaker 4] (2:26:30 - 2:26:31)

Hospital? Why would the hospital pay?

[Speaker 3] (2:26:31 - 2:26:35)

To get better ranking doctors and to have quality check of their own doctors.

[Speaker 4] (2:26:36 - 2:26:43)

Yeah, that's what I was about to say. Repetition for hospitals. Hospital, then it becomes like...

What's this site?

[Speaker 3] (2:26:43 - 2:26:44)

Cliquel...

[Speaker 6] (2:26:45 - 2:26:47)

S&P, Moody's...

[Speaker 1] (2:26:48 - 2:27:10)

No, but still, I mean, while it's game to some extent, but it's still reasonably credible. You think about it. There was this company called Second Opinion.

And they found out, they did a survey, and they found out that almost 41% of the surgeries were unnecessary. So how is it that two doctors, one is saying go for a surgery and one is saying don't go for a surgery?

[Speaker 3] (2:27:11 - 2:27:18)

So what we're saying is it's an intermediary quality agency which sits in between a patient and a doctor in a hospital. Right? At a very broad level.

[Speaker 1] (2:27:19 - 2:27:50)

And for at a premium level, a more curated subset of that for people who can afford. I think both are opportunities. Nikhil, the one thing we missed is talking about is devices.

So I was just talking about a device that's available in the US today to measure VO2 max for a normal person. You know, no, it's not a hundred thousand, you know, complicated device. Right?

I think a lot of these will have value in India and good monetization potential too.

[Speaker 4] (2:27:52 - 2:28:18)

And… No, but most of the devices… Yeah.

JC, excuse me on this, but it's really, are not really Indian made. As in you're forced because there is no supply chain of these devices. So you're forced to go to China, pick up something and come and sell it as in…

No, but I mean, let's leave the China angle aside. Right?

[Speaker 1] (2:28:18 - 2:28:24)

So you're as a, you still are owning the IP and you're manufacturing it in China.

[Speaker 4] (2:28:24 - 2:28:34)

Right? So hopefully… But the thing about my whole thing about this devices thing is the guys with the biggest distribution are going to kill it.

Yes.

[Speaker 1] (2:28:34 - 2:28:44)

See, now I'm somewhere taking Nikhil's startup thing as not only startups that can become all 500 billion to a billion, right?

[Speaker 4] (2:28:44 - 2:29:01)

I mean, the thing is, if you're only a wearable company, getting to that itself is really tough because how do you differentiate? Now, JC has got, Fitter has got a community. So he can sell them rings, right?

[Speaker 1] (2:29:01 - 2:29:03)

No, no. Our primary business model is not the ring itself.

[Speaker 4] (2:29:03 - 2:29:03)

Yeah.

[Speaker 1] (2:29:04 - 2:29:09)

You know, the primary business model is changing the metrics. Once you get the data, what do you do with the data?

[Speaker 4] (2:29:09 - 2:29:24)

So, but how do you, how do you, the thing is wearable. Now, if this person goes and gets the VO2 max, whatever device, there'll be 10 others replicating it in no time. No, no.

There is IP.

[Speaker 1] (2:29:24 - 2:30:04)

That's what I'm saying. See, some of these devices, I believe in India, in the next 10 years, there will be some IP. Every IP need not be just a follow on IP, like what we've had in the last 10 years

is where I'm going with it, right?

Why can't we believe that there is certain medical device IPs that will be novel and at a price point in India that will be more usable and deployable in the country compared to just importing that stuff.

[Speaker 4] (2:30:04 - 2:30:10)

If an Indian can come up with the IP that he can call his own. Yes.

[Speaker 1] (2:30:10 - 2:31:16)

No, Nikhil, but see, even if you make any high-end manufacturing, take any high-end device, 90% of all electronics is China, Taiwan, and South Korea. So you cannot truly control the IP from a hardware point of view. But there's a lot of IP involved in the not hardware side of things.

For example, you can patent on the design. All right? You can patent in the algorithms.

You can patent in the function. So there's a lot of IP-able thing that you can do. But to the point where, nowadays, anybody can challenge any IP and all you have to do is just make small differentiation and then it'll take years for somebody to prove that it's the same thing.

So even IP is not entirely protectable. So I think in today's market, the thing is first-to-market distribution. That's pretty much it.

So… And you can also look at distribution. I mean, again, just to look at UltraHuman, right?

They went to every country where Aura is not there first and said, can we solve for…

[Speaker 4] (2:31:16 - 2:31:18)

No, UltraHuman has done a killer job. Yeah.

[Speaker 1] (2:31:18 - 2:31:44)

So I'm saying, why can't we have 10 such companies come up? You know, people will learn from the old playbooks. We'll build on those playbooks.

And like I said, in science and engineering, I think we are getting better. You know, I think UltraHuman also made at least a few at a software level, I think. They did some innovations.

No, it's never just the hardware.

[Speaker 2] (2:31:44 - 2:31:45)

Yeah, trust me.

[Speaker 1] (2:31:45 - 2:32:09)

When you get the hardware, the hardware is like bare minimum. So you can take the hardware, 10 other people can take the hardware. Design is where you have elementary control.

Like, okay, my device is going to look a certain way. It's going to feel a certain way. Software is where the differentiation lies.

Because at the end of the day, what is the user looking at? He's looking at the data on his phone. What all features are there?

[Speaker 3] (2:32:09 - 2:32:21)

How accurate is it? Like, if you extrapolate, everything is China and TSMC, right? Including Apple and Nvidia.

But the AI and the software layer on top of it, right, will be something that will make a difference and you can own.

[Speaker 1] (2:32:22 - 2:32:23)

This is, I guess, what we're saying.

[Speaker 4] (2:32:23 - 2:32:45)

No, I'm not talking about the manufacturing. I'm talking about Apple going to China saying, I want to make Apple Watches, versus someone else going to China saying, I want Apple Watch. Right?

As in, it's two different things. But let's look at the idea of variables itself, right?

[Speaker 1] (2:32:45 - 2:32:56)

In 10 years, I think we'll be in a different place. The type of things we'll be sensing and measuring continuously, I think will be dramatically different.

[Speaker 4] (2:32:57 - 2:33:11)

And there, I believe there are a lot of opportunities for Indian entrepreneurs to... One of the things I hope is a chip somewhere planted in between. It can measure my glucose.

It can measure my... I'm sure that...

[Speaker 1] (2:33:11 - 2:34:09)

So, your cortisol, your glucose, and... Blood report, everything. Dynamically, right?

But my guess, 10 years. Yeah, I would love for someone not to draw my blood. There are some countries which are doing research on this.

Yeah. They just plant a chip inside your body, they'll do a surgery, and then it'll measure everything. There is one device now that can measure lactate and glucose continuously.

So, you want to know your result? Like, install, as in... No, no, no.

It's just senses. It senses at the skin level. I mean, there are certain skin level changes that you can sense when your lactate levels are higher, right?

So, there's some of these devices for... Because how do you know you're in zone 2? Through some lactate measurements.

So, you're...

[Speaker 3] (2:34:09 - 2:34:17)

Popular opinion, if you had to buy one device, poop, or an ultra-human, Apple, Google...

[Speaker 6] (2:34:18 - 2:34:18)

Yeah, I'll go like this.

[Speaker 3] (2:34:19 - 2:34:24)

JC Printer. What device do we buy? One device.

Don't say anything.

[Speaker 4] (2:34:24 - 2:34:29)

I know your answer. But I'll ask you. It's a heart ring.

[Speaker 1] (2:34:30 - 2:34:35)

I might be the only one who'd be able to explain you what each of these things do. No, no. Please go ahead.

[Speaker 6] (2:34:35 - 2:34:36)

Yeah, yeah, go for it.

[Speaker 1] (2:34:36 - 2:37:01)

Help you improve them. Because most of the people who buy these gadgets, right? They don't know what to do next.

I think that's... And that actually is the problem we were trying to solve. We didn't want to be like another gadget.

First, we sent out a proposal to these companies. And we were like, Hey, we have a community. We have coaches.

We are amazing at helping people fix their lifestyles. Why don't you guys partner with us? And when we didn't see any response, then we looked at, Can we do something of our own?

So it was like watches, bands. And they were like, Okay, ring is the best. Because it technically has the more number of hours of people wearing them.

Because watches tend to get uncomfortable. And so today, our differentiation is not the hardware. The differentiation is look, here's the hardware, which is decently good.

It provides all the metrics as accurately as any other devices. Probably better than smartwatches. But you look at the metrics.

And what next? Right? Which is where you keep wondering.

If you are with somebody else, you'll be like, Hey, this is my VO2 max. Now, what the hell is VO2 max? Or this is my RHR.

What the hell does my RHR indicate? What is this HRV fluctuation happening? What should I measure?

Right? So even though there's a lot of data on what these are, there's no data on how can you systematically get them better. So it's not about the device.

It's what sits on top of the device. But Jassi, things like, let's say, how your activity is measured, right? Are there some such functions where one device is better than the other?

Yes. So it all depends on sampling, right? For example, again, you have Apple Watch, for instance.

They don't sample your heart rate as frequently as, let's say, a Whoop. So in terms of sampling, Whoop has the highest sampling rate. And that's why it's most accurate.

So when we do calculation for certain things, like for example, HR, SPO2 and HRV, these are pretty standard functions. Most of the devices will get them accurately. Right?

But when we talk about, let's say, calorie expenditure. And activities, I think that Whoop kind of... So Whoop would be far more accurate because it's sampling the heart rate every minute.

And then because of that, it's using a better formula, which is your heart rate load, because

there's a direct correlation between heart rate load and your energy expenditure. Other devices are using MET equivalents. You know the problem with MET equivalents, right?

They are a generalized derivation of... from a generalized set of people. So they're not very accurate.

[Speaker 4] (2:37:01 - 2:37:12)

So you go to the gym, your Apple Watch will show you you've burned 600 calories in an hour, whereas a Whoop or a heart rate will show you only burned 250 calories. But what I've seen... Okay, I've used everything.

[Speaker 6] (2:37:12 - 2:37:12)

Yeah.

[Speaker 4] (2:37:13 - 2:37:37)

I mean, I find Apple Watch convenient. And the reason I find it convenient, it's not very intrusive. About this calorie thing, I get it.

But the thing is, it's consistent along the lines. So if you know what is burning 1000 calories on Apple Watch, you can assume it's not 1000, it's 600, 500, whatever. Right?

But then... But it gives you a consistent 500. Now...

[Speaker 5] (2:37:37 - 2:37:38)

So does Whoop, right?

[Speaker 4] (2:37:39 - 2:37:43)

Yeah, it does. So consistently, it does. But the thing is, Whoop is wearing...

[Speaker 5] (2:37:43 - 2:37:44)

You have to wear it 24 hours.

[Speaker 4] (2:37:44 - 2:37:45)

24 hours is...

[Speaker 5] (2:37:45 - 2:37:46)

I wear it on my feet.

[Speaker 3] (2:37:47 - 2:37:55)

I used to wear it on my feet for three months. But then you have to carry that separate... I think every device in the world should have a C-Type.

[Speaker 4] (2:37:55 - 2:38:00)

Yeah. It makes no sense to have a rear charger for a rear C-Type. We added a C-Type.

[Speaker 3] (2:38:02 - 2:38:04)

Yeah, I mean... That's why I stopped wearing it.

[Speaker 1] (2:38:04 - 2:38:24)

What about CGM? CGM, again, I'm not a huge fan of CGM. It's great for diabetics.

But in non-diabetics, there's no data that suggests it helps non-diabetics. I mean, apart from making people feel scared of every single food on the planet that spikes their blood glucose. And you know that blood glucose spikes are completely fine.

[Speaker 2] (2:38:25 - 2:38:26)

There's no problem with blood glucose spikes.

[Speaker 1] (2:38:26 - 2:39:16)

They need to come down in a reasonable way. Yeah, exactly. So...

But for somebody who is probably, you know, in some stages of pre-diabetes, right? And wants to really optimize the type of food they eat. Yeah.

And optimize maybe also the eating windows and kind of go through that phase of seeing what works for them. Yeah. I think, in my opinion, it has value.

There's a level of personalization that the devices can offer. Yeah. But I think today that level of personalization is not like a hidden treasure, I would say.

Everybody would agree, hey, eat a little bit earlier than your bedtime. Eat fewer meals than you would. Like don't hop on six or seven meals.

[Speaker 2] (2:39:16 - 2:39:16)

Right.

[Speaker 1] (2:39:17 - 2:39:38)

Then eat within your calorie intake. If you want to figure out that basis a CGM device, sure. But I'd

say that you're kind of overcomplicating stuff.

Because this stuff is out there in the open. You know, but if somebody feels that, you know, instead of nine o'clock, my CGM tells me to eat at eight o'clock, more power to you.

[Speaker 3] (2:39:40 - 2:39:57)

Longevity supplements. This is the big one. Everybody will.

I know we're all taking different things. Even though JC might say no, I'm not taking anything. I can go on record and tell you guys I'm not taking anything.

I was looking through a bag and I found this growth hormone.

[Speaker 1] (2:40:08 - 2:40:18)

So I think supplements will continue to be a huge business. Yeah. Unfortunately, yes, it's going to be a huge business.

What should one do?

[Speaker 6] (2:40:18 - 2:40:19)

I get Brian's logic.

[Speaker 1] (2:40:19 - 2:40:25)

It makes sense. But again, I'm going back to, you know, JC would agree that you have to measure, right?

[Speaker 3] (2:40:26 - 2:40:28)

And without measuring, what would you recommend?

[Speaker 1] (2:40:29 - 2:40:31)

Vitamin D. No, no, no.

[Speaker 3] (2:40:31 - 2:40:34)

Based on the generic data set of Indian males.

[Speaker 1] (2:40:34 - 2:41:00)

Based on generic. I think most of us have this vitamin D plus K deficiency, right? I mean, that combination, you can't go wrong at a broad level.

Omega 3? Omega 3 also for most of us. Magnesium?

But again, omega 3, I feel you should measure. I think at some point we need to be able to really measure and say, what is good? I mean, how many units of omega C?

[Speaker 3] (2:41:00 - 2:41:01)

I have two capsules.

[Speaker 1] (2:41:01 - 2:41:08)

So I don't know. Why is two good enough, right? Maybe it should be one.

Maybe it should be five. So I think… Five is too much.

[Speaker 3] (2:41:09 - 2:41:09)

One is too little.

[Speaker 1] (2:41:12 - 2:41:27)

So we're all, we're like guesstimating and we're taking some broad… But magnesium seems to be something that's… Everybody agrees on magnesium.

Yeah. I mean, it does help most of… Because the number of…

I was talking about…

[Speaker 3] (2:41:27 - 2:41:30)

Give us the fancy stuff, Prashant. I know you take some fancy stuff.

[Speaker 1] (2:41:31 - 2:42:04)

See, I think I've found that there's enough data on lutein and zeaxanthin for your eyes. What? Lutein and zeaxanthin.

We just don't get enough of it in our diets. It's for your eyes. For your eyes.

And once you're in the 40s and… You take it every day. Take it every day.

And there's enough… I mean, research data on these specific ones, unlike NMN and some of these other things we discussed.

[Speaker 3] (2:42:04 - 2:42:05)

NMN you take every day?

[Speaker 1] (2:42:05 - 2:42:38)

No, I don't take… NAC you take? I take…

See, I take creatine. I think creatine, if you are exercising and, again, widely studied, enough data on creatine. Enough data.

And if you are 50 plus, I think it is beneficial. You should do creatine. Right?

So, I'm telling you things that I've… I know that there is data on. Right?

I like Prashant. There is no data on collagen.

[Speaker 3] (2:42:38 - 2:42:41)

He says no supplements. Everything you're saying is… No, no, no.

[Speaker 1] (2:42:41 - 2:42:45)

I mean, like most of the longevity supplements. These are not longevity supplements.

[Speaker 2] (2:42:45 - 2:42:45)

These are, yeah.

[Speaker 1] (2:42:45 - 2:42:50)

So, NAC, I think there is some data.

[Speaker 3] (2:42:50 - 2:42:59)

NAC is precursor to glutathione. Yeah, glutathione. I mean, if…

I take glutathione IV maybe once in a month along with… There's not enough data. Vitamins.

[Speaker 1] (2:42:59 - 2:43:27)

I take something called Megaboost. Not enough data on just glutathione. So, there is some decent data on astaxanthin.

Astra? Astaxanthin, which is for your skin. Right?

It's an antioxidant. But it seems to have enough data. I mean, I would say in the data…

Oral supplement. Like if creatine and some of these are like 9 on 10, I would say this is like 7 on 10 in terms of…

[Speaker 3] (2:43:27 - 2:43:43)

You know the thing with IV? Yeah. It feels like you're doing something.

Like popping a tablet doesn't feel enough. When somebody comes, pokes a hole in you, and there's a bag, and you watch, drop, drop, drop. You feel like you've done something.

[Speaker 1] (2:43:43 - 2:43:53)

So, the one funny supplement set, Nootropics, right? Like Ashwagandha. I found zero use for myself.

Right? You're going to get a lot of hate for this, by the way.

[Speaker 2] (2:43:53 - 2:43:54)

I know, I know.

[Speaker 1] (2:43:54 - 2:44:16)

Let them… I mean, that's what I'm saying. Ashwagandha, and this I go back to saying, where every supplement is not for everyone.

Right? It's not like Ashwagandha. I mean, just because Huberman says Ashwagandha is good, doesn't mean that everybody…

If it works for you, yeah. Not even Shilajit? No, I don't take Shilajit.

[Speaker 3] (2:44:16 - 2:44:21)

I have people who have been in this house earlier today, swear by it. Practical experience.

[Speaker 1] (2:44:21 - 2:44:30)

I mean, if it works for them, that's great. Right? But I'm not using Shilajit.

All Shilajit studies are funded by the company that makes Shilajit formulation. PrimaVie.

[Speaker 3] (2:44:31 - 2:44:34)

Don't they say they find it on some rock in the mountain?

[Speaker 1] (2:44:35 - 2:44:40)

I mean, all you have to ask them is, what exactly is there in Shilajit? Fulvic acid.

[Speaker 6] (2:44:40 - 2:44:41)

Yes.

[Speaker 1] (2:44:41 - 2:44:52)

And what part of the HPTA access is it helping? Is it helping your hypothalamus? Is it helping your luteinizing hormone?

Is it affecting your FSH? How exactly?

[Speaker 2] (2:44:52 - 2:44:53)

There's no known mechanism.

[Speaker 1] (2:44:53 - 2:44:58)

I think in some of this, the mechanisms itself is the problem, right? We don't know the mechanisms.

[Speaker 6] (2:44:58 - 2:44:59)

So, what's with the collagen?

[Speaker 1] (2:45:00 - 2:46:25)

See collagen, there's like, I mean, most studies again have failed. Very few collagen studies have succeeded. I like Prashant more and more.

Yeah. And there is, you know, it's a very poor protein. If you're taking it to complement your protein, will it do any harm for you?

I don't think so. You can spend money on it and be happy. But it's not going to do much.

Right now, there's not enough. Collagen production is also gene regulated. So, you taking oral collagen is not going to be the same as you taking…

Because it's getting, in your gut, it's getting… Yeah. Because you taking creatine, now creatine has specific receptors that help creatine, you know, go through the cell membrane.

So, CRT receptors. But collagen, if you just take collagen, it'll end up in your bloodstream, but it will not affect any of these things. It's not getting into any cell membrane.

It's not. And what you are getting is basically essential amino acids, which your body can automatically produce when in times of need. So, you taking collagen supplementation is not going to, in any way, any way meaningfully, increase the production of collagen.

That's completely determined by collagen-specific genes, which again are determined based on

your lifestyle and, you know, other epigenetic factors.

[Speaker 3] (2:46:26 - 2:46:26)

What else?

[Speaker 1] (2:46:28 - 2:46:41)

What else? I think top of the stack, I think these are the best. I do do some postbiotics, prebiotics, inulin.

[Speaker 3] (2:46:41 - 2:46:44)

You were telling me the other day prebiotics are as important as…

[Speaker 1] (2:46:44 - 2:48:23)

Yeah, prebiotics are as important, right? So, and I think there, it depends on how much fiber you're otherwise consuming. Do you want to supplement some fiber if you just can't eat that much fiber?

But it just depends on your nutritional value. But I think it's a slightly more complex thing, this prebiotics, postbiotics. But I think it's something that I would definitely put some effort on.

I think here, I'd just like to flag that, and this, I mean, I went through this. Because of the tropical environment we are in and all kinds of food that we eat, the yeast in our gut is a very common thing among Indians. And we all, I mean, how do we fix it?

We keep taking antacids and we think that's the way to solve our gut problem. But actually, you got to, it's called candida, right? I don't know if you've heard of candida.

Right, it's a type of yeast and very, very common in all of us. If so, the extra bloating that you feel, all of it is because of this candida yeast. And there's a little bit of that in everybody.

But when it overtakes and it also compromises your membrane, your gut membrane, it kind of leads to other complications. But I think I would just not worry about just general gut micro...

[Speaker 3] (2:48:23 - 2:48:26)

So what do you do for this candida?

[Speaker 1] (2:48:26 - 2:48:47)

Candida. What do you do for it? You need to go to a good, I mean, there are a few places in India that will, of course, at some point, many more places.

But getting a credible gut microbiome screening place is also difficult in India. But if you find a good one, then they'll be able to detect candida.

[Speaker 3] (2:48:48 - 2:49:02)

Also to everyone, full disclosure, we are not asking anybody to take any of what we are eating. We are just talking about the tablets and supplements we are eating. Nobody listen and go eat any of what we say.

No, definitely.

[Speaker 1] (2:49:03 - 2:49:18)

So the other question is, when it comes to omega supplements, right? The algae versus fish oil. I constantly go back and forth.

Krill oil. Huh? Krill oil.

Oh, krill oil.

[Speaker 5] (2:49:19 - 2:49:20)

I don't know. Krill is a fish, right?

[Speaker 1] (2:49:21 - 2:49:23)

No, krill is not a fish. Yeah, it is. It is?

[Speaker 5] (2:49:23 - 2:49:24)

I thought, yeah, krill oil.

[Speaker 1] (2:49:24 - 2:49:41)

It is a fish oil. So I think algae is the only plant-based omega supplement. Not enough data or as much data as fish oil.

And fish oil, everybody says, gets rancid. You know, it's going to…

[Speaker 3] (2:49:41 - 2:49:43)

Not if you have it right after food.

[Speaker 1] (2:49:44 - 2:49:48)

No, no, no. Rancid as in the… See, one is the source of the fish.

[Speaker 3] (2:49:48 - 2:49:49)

Inside your body or?

[Speaker 1] (2:49:49 - 2:50:07)

No, no, outside. Outside. You don't know if what you're eating is already rancid.

And you don't know the source of that and how much mercury was there in that fish, right? So it's a little bit of… I'm like always still struggling with finding the right…

So I think somebody for India.

[Speaker 3] (2:50:08 - 2:50:17)

Is there a brand of supplements you trust? Like, I feel like there's a huge opportunity here for somebody to display the testing and give you the supplements.

[Speaker 1] (2:50:17 - 2:50:41)

I think Brian was speaking about that when it comes to food, right? And me and Nitin were also talking about it recently. I think there is a huge opportunity for a brand that can be out and open on the lab testing they have done.

And the problem is, what is that credible lab that you will base your decision on?

[Speaker 4] (2:50:41 - 2:51:17)

First thing any company should do is get a spectrometer. Like every… I mean, you can't…

You have to get lab in-house. Because as long as you're getting tested… You mean at home you're saying?

No, in-house as in the… Distribution, the distribution. Because you want to be testing at a certain frequency and you can't do it if you send it outside.

And not just with supplements, even with food. As in, just to be… Like, I question every single thing that's put on…

Do you have a spectrometer at home? Spectrometer is expensive. No, but supplementation…

[Speaker 5] (2:51:17 - 2:51:19)

He wants to now start growing his own vegetables, so…

[Speaker 1] (2:51:20 - 2:51:24)

No, but supplements in India, I think, huge question. Yeah.

[Speaker 2] (2:51:25 - 2:51:25)

So…

[Speaker 1] (2:51:25 - 2:51:51)

They don't come from… Like, they're not engineering. They're not like home-bred.

Because even if we talk about whey protein, it's not that we have whey protein companies in India. You know, like, besides Amul, I don't think… They're just importing.

We are importing everything. And when it comes to whey protein, we're just mixing stuff, labeling it and selling it. And most of the manufacturing happens by the same set of guys.

[Speaker 6] (2:51:51 - 2:51:54)

So, yeah, that's also a problem.

[Speaker 1] (2:51:54 - 2:51:54)

Okay.

[Speaker 3] (2:51:54 - 2:51:57)

What about the tablet you were asking, Brian, about?

[Speaker 1] (2:51:58 - 2:52:01)

Rapamycin. Yeah, see, rapamycin. Now, let's come to the longevity.

[Speaker 3] (2:52:02 - 2:52:04)

No, let's, I think, finish off this supplementing.

[Speaker 1] (2:52:05 - 2:54:24)

Yeah, the rapamycin is one supplement. I think people take. I think…

Are you on it? I'm not on it. Because I think Longevity India is going to do a trial on this in India on rapamycin.

You take it once a week. And it has shown to, you know, modify or act on the mTOR pathway. And that mTOR pathway is the pathway that, you know, in a lot of clinical trials with mice and with other mammals has shown to be most effective to increase longevity.

Growth hormones? Not very. Even for rapamycin, see, what happens is like…

And so, he's talking about mTOR pathway. So, it's a mammalian target of rapamycin. So, there's basically two different pathways in body.

So, let's say you have states of high energy availability and high amino acid in circulation. AMPK and your… Yes.

So, you have the AMPK and mTOR pathway. Now, there was a research last year. This guy came out and he said a higher mTOR activation because of leucine resulted in increase in macrophage activity.

You must have read that, right? And so, he wasn't implying anything. And they did this testing in rats.

It was a mechanistic study. And then you had this whole army of influencers going up and saying protein is killing you. Because it's activating mTOR pathway.

By the way, so, rapamycin in oral form would, let's say, affect your mTOR pathway. The same can be also affected by simply working out, eating enough amount of protein. So, if somebody is not working out and not eating enough amount of protein, they can benefit from taking rapamycin.

But if you are working out and you're taking adequate amount of protein, no difference. So, and the AMPK pathway is the one that, you know, is a pathway that gets activated when you fast for certain periods of time. So, that's the pathway that impacts, you know, cleanup and…

[Speaker 3] (2:54:24 - 2:54:25)

Fasting is good as a no-brainer, right?

[Speaker 1] (2:54:25 - 2:55:43)

Everybody agrees. So, there, I think, there is, I mean, I'm starting to, like, look at not just time-restricted eating, which is one, which is the most common thing. But real fasting, which is, let's say, this 3 plus 2 day modality, where 3 days you just eat the normal amount of calories.

Or maybe you can even increase it a little bit, so your weekly averages don't change. And 2 days of maybe 50% of the calories, and you cycle this. Basically, average out the calories.

Average out the calories, but do this kind of… That seems to have… There seems to be more and more data I'm getting convinced on in doing this 3 plus 2.

I'm doing 16-17 hours every day. That is on a day, that's fine. But then I'm saying that's not fasting.

Do you agree? I mean, see, fasting itself as a first principle is that fasting puts you in a state of calorie deficit. We have lots of studies.

We have the calorie 1, calorie 2, the whole calorie trials. And then we also have lots of studies which have proven that calorie deficit, a slight calorie deficit, over a period of time results in improved mitochondrial efficiency and improved mitochondrial efficiency.

[Speaker 2] (2:55:43 - 2:55:45)

The overall body markers improve. Yes, they improve.

[Speaker 1] (2:55:45 - 2:57:02)

Across the board. So, as long as fasting is getting into a state of calorie deficit in a sustainable manner, it's good. But for a lot of people, like people who go and drive fast, for example, or let's say a three-day continuous fast without eating anything.

Yeah, those are… So, periods of extreme eating and periods of extreme starvation. So, you'll have mTOR pathway and then you'll have simultaneously AMPK.

But both result in extremes, right? And that's not sustainable. That's not good.

So, as long as it's sustainable, fasting can be a good way for somebody to achieve caloric restriction. Growth hormones? Not very…

I don't have very… Yeah, I don't support them because growth hormones, like anything that you put in your body in an exogenous manner and takes your… What if your body has a deficit?

If you're measuring it, if you have a clinical diagnosis that you don't have enough, let's say, human growth hormone, then in that case, yeah, sure, it makes sense. They give it to a lot of kids who have deficiencies when growing up to increase their height and other things. But both testosterone and growth hormones, I think there are ways in which you can organically increase it through a set of things.

[Speaker 4] (2:57:02 - 2:57:33)

I mean, not just supplementation, but a set of things that you can do. For growth hormone, that insulin-like growth factors, IGF-1, it's a certain range. And if you're at the lower end of the range, you can optimize to be at the higher end.

You have to keep taking it. No, but growth hormone is not required. Testosterone, you have to be taking it.

Growth hormone… There's a certain cycle you still have to follow for growth hormones, right? You need to do a certain cycle and then stop it.

[Speaker 1] (2:57:34 - 2:58:17)

And the problem with growth hormones or any hormones, for example, they can penetrate the nucleus. And again, the same set of side effects exist. So, I mean, most of the growth hormones, they are isolated.

So, they're pretty safe. But you cannot absolutely guarantee that they are going to target specific cells. So, one of the side effects of growth hormones is growth of other organs, which has unintended consequences.

Let's say you take growth hormones to improve your muscle hypertrophy. Heart is also a muscle. So, people observe megaly in their heart, right?

So, hearts grow bigger. And so, organ enlargement is also a common side effect of growth hormones.

[Speaker 5] (2:58:17 - 2:58:23)

So, is this anywhere related to hormone replacement therapy? HRT is different.

[Speaker 1] (2:58:23 - 2:59:46)

HRT is different. Growth hormones are different. But if you're talking in HRT, so maybe TRT, and then probably for women, they prescribe HRT.

If you are lacking in them, or if you generally chronically have low levels of, let's say, progesterone, or in case of males, it would be testosterone. So, in which case they can prescribe. But there has to be a clinical diagnosis.

And there should be a good enough reason. Because these things come with their own side effects. But there's a little bit more data on HRT being useful in certain specific situations.

So, I think… Testosterone? Same.

Exogenous testosterone. First of all, it shuts down your natural testosterone, right? So, if you are willing to go through testosterone replacement therapy every year, I think if it works for you, sure, it can provide you vitality.

It can make you feel great. But what have you seen to kind of increase it? It will increase your cardiovascular risk.

And how do you take it? Oral supplement? No, no.

Hormones need to be injected. Like most of these hormones will have to be injected. Orally, they will not be like…

[Speaker 3] (2:59:46 - 2:59:58)

All this tonkat ali and all of that nonsense? No, no data. Cordyceps.

They don't work. Some of these huberman types talk about it. Tonkat Ali does.

[Speaker 4] (2:59:59 - 3:00:04)

Tongkat Ali and Fudoshia are just this. So, Huberman also has a conflict of interest.

[Speaker 1] (3:00:05 - 3:00:11)

Because AGI promises to have all of these things. And Huberman has an investment in AGI. No, no, no.

[Speaker 4] (3:00:11 - 3:00:54)

AG1 doesn't really have this. Tongkat Ali and Fudoshia, it doesn't. AG1 is essentially a vitamin and a mineral drink.

That's like the green juice, which has all the micronutrients. It's more micronutrients. I mean, the problem Brian had was the fact that they're paying influencers quite a bit.

So, if you're selling something for $100 and you're paying $20 forever to the influencer… But then Brian's stuff is also very expensive. It's not cheap.

So, that's what I wanted to ask. Yeah, it's very expensive. But then he said that what goes into it is open sourced.

[Speaker 3] (3:00:54 - 3:00:55)

Isn't AG1 like that?

[Speaker 1] (3:00:56 - 3:01:03)

No, no. AG1 has this other problem, right? They don't tell you what exactly is there.

I mean, that is a problem.

[Speaker 3] (3:01:04 - 3:01:08)

So, final takeaways. I think we've recorded for near five hours.

[Speaker 1] (3:01:09 - 3:01:15)

Four hours. Nusair wasn't joking when he told me that the podcast went on for five hours.

[Speaker 3] (3:01:17 - 3:01:25)

It's short. Takeaways in the sense, A, outside of the supplements you mentioned, first thing I want to know, is there something you're taking that he's not?

[Speaker 1] (3:01:26 - 3:01:38)

No, I'm only taking vitamin B12. And I'm taking Finasteride. Finasteride is?

It's for hair, basically. So, if you have A, which is androgenic alopecia. So, male pattern.

[Speaker 3] (3:01:39 - 3:01:42)

That's actually an interesting question. Is there something for hair to reverse balding?

[Speaker 1] (3:01:42 - 3:02:44)

No, not reverse. Finasteride does what? Basically, it's a DHT blocker.

Minoxidil? Minoxidil is more like caffeine. You know, it will stimulate hair follicles.

So, it will stimulate hair follicles. It will give you, but again, the amount of alcohol they put in minoxidil, it's likely not good for you. So, I'd rather stick to something which is more caffeine-based than minoxidil.

So, the B supplement, sorry, I missed it. I do take some B supplements. Vitamin B is like B complex?

No, it depends on which. You don't need necessarily all of them. So, you shouldn't be taking a B complex.

Whether it's B6 or B12 or folate, you know, which depends on. So, in my blood test, I do these like every quarter. So, in that, I had a deficiency of B12.

So, again, yeah. And I think you get it otherwise from animal sources, right? Meat, B12 is from?

I didn't check the sources.

[Speaker 3] (3:02:44 - 3:02:45)

I forget.

[Speaker 1] (3:02:46 - 3:02:52)

But I think B12 deficiency is a common one. So, I think it was a big miss that I didn't mention that.

[Speaker 3] (3:02:52 - 3:03:06)

But today when JC was coming and we were asking everybody what you'll eat for dinner, like, don't tell anyone, but I want a steak. Nobody knows. Who, me?

You know, this is amazing.

[Speaker 1] (3:03:06 - 3:03:08)

I'm learning things about you. I didn't know myself.

[Speaker 3] (3:03:08 - 3:03:09)

You don't eat eggs apparently, right?

[Speaker 1] (3:03:09 - 3:03:15)

No, but steak is one of the proteins. You don't eat anything but eggs, no? No, I had salad, I had food.

[Speaker 3] (3:03:15 - 3:03:26)

No, no, but like in meat? I had chicken. I thought you were vegetarian.

Hence, I was making this. I was. Tonight, I had to.

Like, in the middle of the podcast, I'm not going to tell someone, hey, I can't eat chicken.

[Speaker 1] (3:03:26 - 3:03:27)

He's not eating that.

[Speaker 3] (3:03:27 - 3:03:32)

Everybody was eating that. What supplements are you guys taking that is not on the list? Yeah, what are you taking?

[Speaker 4] (3:03:33 - 3:03:37)

I'm on a whole ton of them. So, I…

[Speaker 1] (3:03:37 - 3:03:44)

I just meant, I'm kind of curious. One or two, which I didn't mention, right?

[Speaker 4] (3:03:44 - 3:04:02)

No, it's… Like, Tongatali, I heard. No, no, I mean, not Tongatali.

But, say, as soon as I get up in the morning, I have thyroid tablets. What? Thyroid.

Thyroid? Okay. Thyroid pills.

Both of those. So, that's more… Then, athletic greens.

Sorry? Athletic greens. AG1.

[Speaker 6] (3:04:03 - 3:04:04)

AG1. That's the whole one.

[Speaker 4] (3:04:05 - 3:04:10)

Then, after breakfast, I take a whole lot of them.

[Speaker 1] (3:04:10 - 3:04:13)

So, like, which are the ones… I mean, I just am curious.

[Speaker 4] (3:04:13 - 3:04:13)

Okay.

[Speaker 1] (3:04:14 - 3:04:15)

Which are the ones that we didn't talk about?

[Speaker 4] (3:04:18 - 3:04:30)

Choline, Alpha-GPC, L-Theanine. Oh, Alpha, I mean… Alpha-GPC.

I mean, this is for mental health. I mean, I… Okay.

I started a lot of this… So, Choline and… Okay.

[Speaker 6] (3:04:30 - 3:04:31)

After the…

[Speaker 4] (3:04:31 - 3:04:53)

You know, after this year's stroke, I thought… I mean, Choline, I've heard. And B12, B2.

Then, yeah, there are a bunch of them. So, I don't know how much it's helping me, but I'm taking it. Seema, what are the ones which we should take?

[Speaker 5] (3:04:53 - 3:04:57)

Nothing that wasn't mentioned here.

[Speaker 4] (3:04:57 - 3:04:58)

Okay. Thyroid is something.

[Speaker 5] (3:04:59 - 3:05:00)

Thyroid is a medical… Yeah.

[Speaker 1] (3:05:01 - 3:07:01)

What do you think of, you know, some of these autophagy supplements, like quercetin? I mean, that's practically what is there in AMPK, right? I know, but if you had to…

Do you think, in addition to what you would do otherwise in fasting and so on, is there enough data on these? There's no data. In fact, fasting is no better than just being in a state of calorie deficit.

We just had a meta-analysis come out a few days. I think the whole myth started when this guy, Yoshimori, he got a Nobel Prize. He got a Nobel Prize for showing pathways of autophagy in yeast, just single-celled.

And people have somehow inferred that autophagy works in humans in a similar manner and we are multicellular, right? You can't just turn it on and turn it off. It's auto-regulated, basically.

So, whenever your body hits the AMPK pathway, and again, both the pathways are active simultaneously. You have the mTOR pathway, you have the AMPK. And the regulation happens depending on states of energy availability.

So, if you are well-fed, if you have enough amount of circulating protein, your body will hit the mTOR pathway if you have done enough exercise, right? It will promote growth and cell proliferation, mitosis. So, if you don't have sufficient energy, then your body will choose the AMPK pathway where it will undergo autophagy.

So, periods of not eating will promote autophagy. Periods of eating will promote mTOR. Extreme of both the phases will disrupt this balance.

So, you should ideally regulate between periods of eating and periods of not eating. And that's basically built into the human physiology. We feel hungry.

Once we eat the food, we are satisfied. If our body wanted extreme amounts of autophagy, we'll have a built-in mechanism where we'll not feel hungry for days. Or we'll keep eating for days.

That's not our body's physiology. It balances both out. So, Nikhil, regenerative medicine.

We didn't talk about that. Okay. Right?

Big area.

[Speaker 3] (3:07:02 - 3:07:03)

Explain regenerative medicine.

[Speaker 1] (3:07:04 - 3:08:11)

Regenerative medicine is, I mean, I guess, the most popular thing in regenerative medicine is stem cells. Right? There is data there.

Very tough to use it. Implementation is very difficult because, you know, you have two problems. One, there is not enough, again, good, validated clinical trials on which type of targeted stem cell therapy works.

Right? I think the most data is on stem cell therapy for joints and knees. I think there is some reasonable amount of data for that.

And for certain other conditions, also, there is data. But India is still, I don't think, from a regulatory perspective, we have everything in place to kind of, you know, operationalize this at a big scale.

[Speaker 3] (3:08:12 - 3:08:15)

India is a good place to experiment with. Not necessarily. No.

[Speaker 1] (3:08:16 - 3:08:24)

Not India. Dubai. Dubai.

Dubai. So, a lot of the stem cell therapies are more popular in Dubai and Cayman.

[Speaker 3] (3:08:25 - 3:08:28)

If I could go to Dubai and get one treatment, what should I get that I can't get here?

[Speaker 1] (3:08:29 - 3:09:03)

I would look at some, I mean, if there are some specific issues you have. No issues, just like that. Just like that.

So, there are certain, the other forms of regenerative medicine, exosomes. Have you heard of exosomes? Exosomes, there are, you know, validated therapies for your skin that seem to work very well.

Right. These are what your cells secrete. So, you just use what they anyway secreted.

Is that, and for hair, I don't know if you have tried.

[Speaker 6] (3:09:03 - 3:09:04)

Yes, yes.

[Speaker 1] (3:09:04 - 3:09:13)

That's probably the only thing that can work for. Yeah. Is this that PRP thing?

No. No. Microneedling.

Microneedling. Yeah. And then followed by exosome.

[Speaker 3] (3:09:14 - 3:09:16)

Does the PRP thing work?

[Speaker 1] (3:09:17 - 3:09:30)

It works. It helps. Because it's basically stimulating your hair follicles and basically nourishing your hair roots with, you know, like plasma-rich blood basically.

That's what PRP is.

[Speaker 3] (3:09:31 - 3:09:44)

I want to ask you like fleeting thought on anything on aging, disease management, aging, disease prevention, therapeutics. No, we discussed that entire thing, right? Targeting age-disease reversal?

[Speaker 1] (3:09:45 - 3:10:30)

Yeah. Age-disease reversal. See, I think, you know, some of these things that you do at a broad level, let's say, you know, whether it's sleep or better nutrition.

I think some of these are reversal at a broad level, right? I think these are more, you know, system-level things that you do too. I think reversing just through one specific intervention, a particular disease may or may not work.

I think a holistic, targeted approach on multiple dimensions is, I think, what works.

[Speaker 3] (3:10:30 - 3:10:42)

Okay. Fleeting thought to our audience of wannabe entrepreneurs. One idea each.

You've invested in a lot of health companies. One thing somebody can start, should start?

[Speaker 4] (3:10:43 - 3:10:52)

No, I think more than the one thing someone should start, it is maybe starting with the right expectation. Because...

[Speaker 3] (3:10:52 - 3:10:54)

They don't want to hear we need to have the right expectation.

[Speaker 4] (3:10:54 - 3:10:59)

No, no, right expectations as in, in a sense, you can't build a billion dollar business in this.

[Speaker 3] (3:10:59 - 3:10:59)

No, no.

[Speaker 6] (3:10:59 - 3:11:01)

I'm not saying billion dollar. Make money business.

[Speaker 4] (3:11:02 - 3:11:47)

Yeah. Make money. I think, I think the food, I am completely, like, like, like we said before, I'm completely sold to the idea that someone is going to put food on the table.

That's, that cannot be questioned. Like right from your staples to vegetables, to fruits, to everything, right? As in to say that this is the food you're eating.

This is the quality from where it was sourced. This is all the pesticide that's there. I mean, and, and provide it clean and fresh.

And I am, I'm sold to the idea. And potentially looking to invest in as many of them as possible. Ashant?

[Speaker 1] (3:11:48 - 3:13:38)

I think medicine is going to change at a very transformative level with all the things that we spoke about, right? So delivering it in a new way is the big opportunity. It's not going to be done by the current medical establishment because they are doing well in what they're doing.

They're very profitable in what they're doing and they don't have a reason to change. And if this is going to be disrupted, it's going to be disrupted by a new set of companies. It's not just biopics.

There'll be many such companies who will find, you know, this whole idea of more systemic, data-driven, leveraging AI, leveraging a more comprehensive approach. I think you'll see a new generation of clinics, hospitals, you know, and I think people will see because the idea of preventive care was always there. So you'll see, you'll ask me, you know, what is so great?

Preventive care, everybody's talking about. But I think there was no data. So just to tell people, okay, eat well, sleep well, just telling them more of that is not going to help.

Can you tell them how can you take action and how can you be specific so that you see change in a reasonable amount of time? And you're able to manage them longitudinally and not in an episodic manner, right? So there is a business model change.

There's an approach change. So there's like a huge opportunity and I think this is like a 25-year opportunity that I see.

[Speaker 5] (3:13:39 - 3:14:00)

Seems mental health. I mean, a lot of people are doing, I don't think, I mean, there is a good place where you know, where people understand what other people are going through. And I think, yeah, that's something help is needed there.

[Speaker 3] (3:14:00 - 3:14:04)

I agree. Some people on this table also need it.

[Speaker 1] (3:14:09 - 3:14:55)

So in future, if India has to make any strides in the medical realm, if it wants to become relevant in healthcare AI, it will need its own data sets. Today, India does not have quality data sets and somebody will have to figure out. Either they create their own personalized labs, work with wearable companies like us, or work with hospitals.

But we need trainable data sets. Without trainable data sets, our health AI is going to be as good as nothing. Like we'll rely on Western data sets, which don't mean nothing.

Because like Prashant was mentioning, the universal BMI doesn't work on Indian population. We have so many Indians who are like BMI is 25, but they have a tummy. So higher visceral fat, right?

So it doesn't work. So you need a lot of data sets.

[Speaker 3] (3:14:55 - 3:14:56)

Okay, I think that's it.

[Speaker 1] (3:14:57 - 3:14:59)

Thank you guys for doing this.

[Speaker 6] (3:15:01 - 3:15:01)

Cheers.

[Speaker 5] (3:15:19 - 3:15:22)

Okay. Hi, Brian. I'm Seema.

[Speaker 2] (3:15:22 - 3:15:23)

Nice to meet you.

[Speaker 5] (3:15:24 - 3:15:55)

I guess so you all have covered about longevity when it comes to men. So my question is, how does it factor when it comes to women, from women's perspective, how longevity factors in when it is regarding women's reproductive health? Or also the bigger question is, how does it factor in menopause?

[Speaker 2] (3:15:55 - 3:15:55)

Yeah.

[Speaker 5] (3:15:56 - 3:15:58)

Do you have any insights on that?

[Speaker 2] (3:15:58 - 3:16:04)

Yeah. What do you do for your health today? Like, let's start with your sleep.

How was your sleep?

[Speaker 5] (3:16:04 - 3:16:05)

My sleep?

[Speaker 2] (3:16:05 - 3:16:05)

Yeah.

[Speaker 5] (3:16:06 - 3:16:08)

Um, I would say it's great.

[Speaker 2] (3:16:08 - 3:16:11)

Great. Okay, cool. Do you measure it?

[Speaker 5] (3:16:12 - 3:16:13)

I do.

[Speaker 2] (3:16:13 - 3:16:18)

Yeah. Great. So like what's on an average night, how many hours of sleep do you get?

[Speaker 5] (3:16:19 - 3:16:21)

Between eight to nine.

[Speaker 2] (3:16:21 - 3:16:23)

Great. And of deep and of REM?

[Speaker 5] (3:16:24 - 3:16:28)

REM sleep, I think around two hours.

[Speaker 2] (3:16:28 - 3:16:29)

Yeah. And deep?

[Speaker 5] (3:16:31 - 3:16:33)

Deep is an hour maybe.

[Speaker 2] (3:16:33 - 3:16:36)

Okay. And then do you get up? Do you find yourself waking up at night?

[Speaker 5] (3:16:37 - 3:16:37)

I do.

[Speaker 2] (3:16:38 - 3:16:40)

Yeah. Once, twice, three times?

[Speaker 5] (3:16:41 - 3:16:42)

Two, three times.

[Speaker 2] (3:16:43 - 3:16:45)

Yeah. Is it just because you're thinking about things or?

[Speaker 5] (3:16:47 - 3:16:51)

No, it's just, I just wake up.

[Speaker 2] (3:16:51 - 3:16:58)

Yeah. Okay. So you're doing pretty well in sleep.

How about diet? How are you doing on nutrition?

[Speaker 5] (3:16:59 - 3:17:00)

Pretty good, I would say.

[Speaker 2] (3:17:01 - 3:17:03)

Yeah. Okay. What does your diet consist of mostly?

[Speaker 5] (3:17:04 - 3:17:13)

1,500 calories. Protein. Healthy carbs, healthy fats.

Okay.

[Speaker 2] (3:17:14 - 3:17:20)

So do you eat much like junk food or? No. Processed food.

Okay. So good. What about exercise?

[Speaker 5] (3:17:21 - 3:17:23)

I get about an hour every day.

[Speaker 2] (3:17:24 - 3:17:26)

Yeah. So what do you do for exercise?

[Speaker 5] (3:17:29 - 3:17:34)

Weight training, there is yoga, cardio. Yeah. Combination of.

[Speaker 2] (3:17:34 - 3:17:40)

Yeah. Great. So you're doing really well.

In the primary areas of sleep, diet, exercise. Good job.

[Speaker 5] (3:17:41 - 3:17:42)

Thank you.

[Speaker 2] (3:17:42 - 3:17:49)

That's very rare. Have you done any testing? Have you had blood draws or have you done MRIs or?

[Speaker 5] (3:17:49 - 3:17:53)

Blood draws, yes. A lot of scans, but no MRIs.

[Speaker 2] (3:17:54 - 3:17:56)

Okay. And the blood draws, when was your most recent blood draw?

[Speaker 5] (3:17:57 - 3:17:57)

Sorry?

[Speaker 2] (3:17:57 - 3:17:59)

When was your most recent blood draw?

[Speaker 5] (3:18:00 - 3:18:01)

Two weeks back.

[Speaker 2] (3:18:01 - 3:18:10)

Okay, great. So you're like an ideal. Yeah, this is amazing.

Okay. And did you discover anything that was an anomaly? Anything that was off?

[Speaker 5] (3:18:11 - 3:18:14)

At present, thyroid levels, I would say.

[Speaker 2] (3:18:15 - 3:18:16)

Iodine levels?

[Speaker 5] (3:18:16 - 3:18:16)

Yeah.

[Speaker 2] (3:18:17 - 3:18:17)

Yeah. Okay.

[Speaker 5] (3:18:17 - 3:18:21)

They're not right. And lipid profile is not great either.

[Speaker 2] (3:18:22 - 3:18:23)

So some cholesterol markers were a little off?

[Speaker 5] (3:18:24 - 3:18:24)

Yes.

[Speaker 2] (3:18:24 - 3:18:25)

Okay.

[Speaker 5] (3:18:26 - 3:18:28)

So I'm thinking maybe it's the vacation food that I had.

[Speaker 2] (3:18:29 - 3:18:40)

Okay. So like it wasn't the whole story of the diet. Okay.

There's more going on. Yeah. Okay.

Yeah. This is usually how it goes, right? So what happened on this vacation?

[Speaker 5] (3:18:41 - 3:18:43)

Well, good food. Yeah. Good wine.

[Speaker 2] (3:18:44 - 3:18:47)

Yeah. Yeah. What kind of food?

[Speaker 5] (3:18:49 - 3:18:53)

What kind of food? I don't know. Whatever Michelin star restaurants cook.

[Speaker 2] (3:18:54 - 3:18:58)

Yeah. Yeah. Yeah.

And the night you had that food and you had the alcohol, how was your sleep?

[Speaker 5] (3:19:00 - 3:19:01)

You know what? Sleep was still good.

[Speaker 2] (3:19:02 - 3:19:02)

Yeah.

[Speaker 5] (3:19:02 - 3:19:05)

I didn't wake up. The way I usually do.

[Speaker 2] (3:19:06 - 3:19:07)

Do you know what the REM and deep were?

[Speaker 5] (3:19:08 - 3:19:10)

I know. I didn't really measure them.

[Speaker 2] (3:19:11 - 3:19:14)

Okay. And so did you exercise the next day? Yes.

[Speaker 5] (3:19:14 - 3:19:15)

Every day.

[Speaker 2] (3:19:15 - 3:19:26)

Okay. Good. Cool.

So yeah, you're doing really well on the core three things and also testing. And so you're checking the box on the most important things and you haven't found any major deficiencies. Is that right?

[Speaker 6] (3:19:27 - 3:19:27)

Yeah.

[Speaker 2] (3:19:28 - 3:19:41)

So as a female, you're doing all the right basics. That's a really great thing. And that's uncommon for a lot of people.

Have you tested yourself for contaminants like heavy metals or microplastics?

[Speaker 5] (3:19:41 - 3:19:42)

No.

[Speaker 2] (3:19:43 - 3:21:24)

So this is one area where you could potentially improve. So a recent study showed that women with high levels of BPA produced substantially fewer eggs and ovulation. That contaminants reduces the body's production of eggs.

And so even when you're doing all the right things with sleep and diet exercise, there are things in the outside world that can come in like contaminants in food, contaminants in the air that can negatively affect your fertility and negatively affect eggs that are produced, negatively affect your hormones. And so there is some opportunity on the measurement side of things that you can try to identify what, like we have a new microplastics test, which we've put out. You prick your finger, you get a measurement and those levels help inform like what is your level of contamination?

So like a simple thing is like in these foods, the question would be what is the level of microplastics in the food? They're there. We just don't know how much.

So we test it. And so a lot of times I do this Q&A with you because a lot of times when you talk about female health and we talk about fertility and menopause, most people want there to be a silver bullet, like, you know, do this, don't do that. It's oftentimes not as clean as that.

Usually it's more nuanced and complex. Yeah. And so I'd say the first thing that's relevant for you as a female is to continue to the basics, sleep, diet, exercise, which you're doing really well, basic testing.

And you can now consider to do more advanced testing to find potential areas of improvement. That'd be like number one. How do you feel about that?

[Speaker 5] (3:21:27 - 3:21:29)

She wants me to ask a question. Yeah.

[Speaker 6] (3:21:31 - 3:21:32)

OK.

[Speaker 5] (3:21:32 - 3:21:47)

Well, if men live forever and women can't give birth beyond 35, what happens to the societal structure?

[Speaker 2] (3:21:48 - 3:22:50)

Yeah. So I think that I'm personally very excited about the future of health. And specifically, I think that AI may produce in the next five years, you know, decades, if not a century worth of progress.

And in those discoveries, maybe something about a woman's time period of fertility. I think, you know, women produce a certain number of eggs over a certain duration of time, and that fertility window closes. And so it's not something that we've solved today, but I don't think it's out of the realm of possibility that we could either extend that or somehow modify that.

But I think it's one of the coolest areas of research of health and wellness is because I agree with you, it's a challenging circumstance that women have this very small window of time. And once it passes, it's gone. Men, you know, have a much longer...

[Speaker 6] (3:22:50 - 3:22:50)

Longer.

[Speaker 2] (3:22:51 - 3:24:07)

Exactly. So I agree with you. So I think it's a really great area of interest.

And so I guess I was asking you these questions because as a female, what you're trying to do is you're trying to slow your speed of aging down as much as possible. Like you want your body to be fertile as long as possible. As long as possible.

All right. And then as you go into menopause, you want your body to be in optimal conditions. And so that's why I focus on the basics.

Because oftentimes people think there's so many differences between men and women, but actually there's so many more things we have in common that we all benefit from sleep. We all benefit from exercise. We all benefit from nutrition.

And then you can dial in specifics like, how are your hormone levels? You know, like what are the deficiencies you're in? What are your contaminants?

So it's the same process as men. You're just simply saying, what is an ideal level for a given thing? How do you measure it?

And then how do you fix it? And so I like to think about that question because... Because it really makes it concrete, right?

And there's a lot of known science on like, how to maintain optimal levels for women. But menopause is just like, that's just a very complicated, challenging thing, just like the fertility window. So sadly, there's not a clean answer there.

It's just, we have to work on it.

[Speaker 5] (3:24:08 - 3:24:09)

One last question.

[Speaker 2] (3:24:09 - 3:24:09)

Yes.

[Speaker 5] (3:24:10 - 3:24:21)

You know, I heard you mentioned that sexual activity overall has gone down. How, you know, to reach that optimal sexual...

[Speaker 2] (3:24:21 - 3:24:23)

Yeah. Yeah.

[Speaker 5] (3:24:23 - 3:24:24)

For all the ages.

[Speaker 2] (3:24:24 - 3:24:57)

Yeah. So did you know that every night when you sleep, you go through several arousal cycles? Where, yep.

No, I didn't. So men, when men sleep, they'll typically have between three and five erections. And each erection lasts, you know, for a certain duration of time.

And so for example, an average 18 year old would have nighttime erections on the order of two hours and 30 minutes. Like thereabouts.

[Speaker 5] (3:24:57 - 3:24:58)

Oh, okay.

[Speaker 2] (3:24:58 - 3:27:25)

Yeah. And so that marker is a representation of sexual, physiological, and cardiovascular function. Now, women have the same arousal cycles with men.

So the clitoris engorges in the same way the penis does through these various cycles, through sleep patterns. And just like men wake up with morning boners, women wake up with morning woods. Oh, I'm sorry, the morning bean.

And so it's the same cycle. And so again, the similarity to men and women. And so this is why your sleep and diet and exercise are so important is because when you're going through these arousal cycles, your sexual function is robust.

And so that would give you a good starting point to be potentially sexually active and to want to engage in sexual intercourse. And so it comes back to the basics every time. Like you do the things that build a foundation of health.

And then you jump to measurement to say, are there any deficiencies you can address? And then you try to fix those. And then you can say, can I optimize further?

So for example, I'll give you one optimization. Okay, so I started, I measured my nighttime erections. There's a little device you put on the penis and it measures the engorgement of the penis and the number of times and the strength of it.

And so my baseline measurement was two hours and 12 minutes. And then I did two therapies to increase my nighttime erections. One was focus shockwave therapy.

So it uses acoustic waves. It feels like an electric shock, but you're basically shocking the penis with acoustic waves. A little painful.

But there's evidence to show it works. People also use it for vaginal rejuvenation. So in the same way where this focus shockwave therapy can be used to take someone who has ED, erectile dysfunction, and bring them back to normal, it can also be used like someone like myself who had a proper sexual function and enhance it.

And so if you're trying to say you have the basics solved and now you want to enhance your sexual function, there are therapies, for example, like focus shockwave therapy, which can enhance the vagina and its overall health. So once you get into the world of this literature, there's this whole world of things you can trial, of things you can do to actually boost your sexual function that are based on scientific evidence.

[Speaker 5] (3:27:26 - 3:27:28)

Ah, perfect. Thank you so much.

[Speaker 2] (3:27:28 - 3:27:28)

What do you think about that?

[Speaker 5] (3:27:29 - 3:27:35)

Great. Well, maybe we all are going to try that.

[Speaker 2] (3:27:37 - 3:27:39)

I think, yeah, it might be an interesting experience.

[Speaker 5] (3:27:39 - 3:27:45)

I mean, why not? So I guess it's sleep time for you.

[Speaker 2] (3:27:45 - 3:27:51)

Yeah. So let me ask you, what are you walking away from this conversation? What did you hear in this conversation?

[Speaker 5] (3:27:52 - 3:28:17)

Longevity is a choice, I guess. I mean, if I'm ready to give up on certain things, be disciplined about certain things, then it's for those people. But if you're not ready to give up on certain things, then maybe it's not so.

I mean, I don't think it's for everyone. Even if you had a choice to participate in it.

[Speaker 2] (3:28:17 - 3:29:39)

Yeah. Well, and I would say, I would say the summary for yourself and others is the basics of health and wellness, sleep, diet, and exercise have the biggest gains of all things. The next biggest gain is finding deficiencies through blood tests, imaging, you know, and contaminants.

And then the third thing is trying to do things to enhance. But I've wanted to try to change the frame of the conversation because I know that women, they think about fertility specifically, and they think about menopause, for example, these two buckets. But oftentimes it really misses the point that they get so much attention that we then forget these basics and we forget the value of measurement.

And so that's why I like the similarities that males and females have is so much of the same process, the same basics, the same measurement, the same evidence. And if we all go through that process, like we're going to hopefully come up with good outcomes. But hopefully that's helpful for everybody.

Whereas instead of specific recommendations for given things, we just go through the same scientific process and we can produce healthy outcomes where you don't really need to choose longevity. You're just choosing health. Like you're choosing happiness.

You're choosing vibrancy. You're choosing good relationships. You're choosing like to be a certain way.

[Speaker 5] (3:29:42 - 3:29:43)

Thank you so much. Yeah.

[Speaker 2] (3:29:43 - 3:29:46)

Is there any other confessions? Like if so...